



Lyceum of the Philippines University – Cavite
College of Engineering, Computer Studies & Architecture

COECSA RESEARCH MANUAL

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2nd Edition 2024

COECSA – RESEARCH AGENDA

RESEARCH CRITERIA

The criteria for research projects shall include:

1. Relevance to the contribution to theoretical and practical knowledge.
2. Support towards the attainment of the Mission, Vision and Core Values of the Lyceum of the Philippines University - Cavite.
3. Contribution to the attainment of national development goals and objectives
4. Relevance to global and international issues with impact on national survival and identity
5. Qualifications and capabilities of the research proponents vis-à-vis the requirements of the research proposal
6. Access of proponents to the logistical support, equipment, and facilities required by the proposed research activities

COLLEGE RESEARCH THRUST

Such research thrusts include the following:

1. Institutional implementation of Engineering, Architecture & Technology to LPU-Cavite in a globally competitive scale;
2. Contribution to the academic discipline through innovative research;
3. Authentic activities engage and motivate students and encourage deep learning—i.e., the development of analytical and cognitive skills—that can be transferred to new contexts; and
4. Collaborative learning or reciprocal teaching where students use one another as teachers, reviewers and critics.

RESEARCH PRIORITY AREAS

Bachelor of Science in Architecture (BSArch)

1. Sustainable Architectural Design Studies
 - a. UN-SDG
 - b. Sustainability and Environmental Science
 - c. Culture and Tourism
 - d. Disaster Risk Management
2. Community Architecture and Urban Design Studies
 - a. UN-SDG
 - b. Sustainability and Environmental Science
 - c. Health, Wellness, and Medical Sciences
 - d. Social Sciences and Human Behavior
 - e. Culture and Tourism
 - f. Artificial Intelligence and Data Science
 - g. Digital Transformation and Society
 - h. Disaster Risk Management
 - i. Social Sciences and Behavior
3. Hospitality Design
 - a. United Nations - Sustainable Development Goals
 - b. Sustainability and Environmental Sciences
 - c. Health, Wellness, and Medical Sciences
 - d. Culture and Tourism
 - e. Disaster Risk Management
 - f. Global and Health Promotion
 - g. ASEAN Studies
4. Architectural History, Theory and Criticism
 - a. UN-SDG
 - b. Culture and Tourism
 - c. Social Sciences and Human Behavior
 - d. ASEAN Studies
 - e. Digital Transformation and Society
 - f. Artificial Intelligence and Data Science

Bachelor of Science in Computer Engineering (BSCpE)

1. Applications of the different fields of specialization:
 - a. Microelectronics
 - b. Embedded Systems
 - c. Software Development
 - d. System and Network Administration
2. Industry-based projects related to computer engineering fields
3. Socio-economic projects related to computer engineering fields

Bachelor of Science in Civil Engineering (BSCE)

1. Alternative Building Materials
2. Innovative Construction Systems
3. Project Management
4. Development of Software for Structural Analysis and Design
5. Maintenance and Management of Large Infrastructures
6. Hazard Mitigation and Infrastructure Renewal related to Natural Disasters
7. Foundation problems in soft terrains
8. Slope Stabilization in High Areas
9. Planning of Transportation Infrastructures in Urban Areas
10. Problems of Flooding and Drainage in Urban Areas
11. Planning for the Mitigation of Natural Disasters
12. Development of Low-cost Sustainable Eco materials for Construction
13. Transportation and Highway Engineering
14. Use of Artificial Intelligence and Machine Learning in Civil Engineering

Bachelor of Science in Mechanical Engineering (BSME)

1. Alternative Sources of Energy/Renewable Energy
2. Mechatronics/Robotics/Nanotechnology/Automation
3. Manufacturing Technologies
4. Thermofluid Science
5. Biomedical Engineering
6. Environmental Technologies

Bachelor of Science in Electronics Engineering (BSECE)

1. Communications
2. Microelectronics
3. Power Electronics
4. Biotech/Biomedical Engineering
5. Instrumentation and Control
6. Information and Computing Technologies

Bachelor of Science in Electrical Engineering (BSEE)

1. Alternative Energy Resources
2. Innovative Electrical Equipment Design
3. Development of Software for Power System Analysis and Design
4. Development of Software for Electrical Circuit Analysis
5. Development of Software for Illumination Engineering Design

6. Design of Means of Transportation Using Electricity
7. Development of Low-cost Sustainable Eco materials for Electrical Installation
8. Other projects related to the practice of the Electrical Engineering profession

Bachelor of Science in Industrial Engineering (BSIE)

1. Production Engineering:
 - a. Development of Manufacturing Processes and Systems
 - b. Development of Product Design and Improvement
2. Organization and Decision System:
 - a. Economic Analysis for Decision Making
 - b. Project Management
3. Ergonomics/Human Factors Engineering
 - a. Methods Engineering
 - b. Work Systems Analysis and Design

Bachelor of Science in Computer Science (BSCS)

1. Current Computer Science Topics
 - a. Software Development Theory
 - b. Mobile Computing Systems
 - c. Software Extensions or Plug-ins
 - d. Expert System and Decision Support System
 - e. Systems Software
 - f. Intelligent System
 - g. Game Development
 - h. Computer Vision
 - i. Image / Signal Processing
 - j. Natural Language Processing
 - k. Pattern Recognition and Data Mining
 - l. Bioinformatics
 - m. Graphics Application
 - n. Cloud Computing
 - o. Parallel Computing
 - p. Embedded Systems
 - q. Emerging Technologies
2. Foundations of Computer Science
 - a. Automata and Formal Languages
 - b. Data Structures and Algorithm Design and Analysis
 - c. Web Semantics
 - d. Coding Theory
 - e. Programming Languages
 - f. Visualization Systems

- g. Computer and Architecture
 - h. Modeling and Simulation
- 3. Human Computer Interaction
 - a. Usability
 - b. Affective Computing
 - c. Emphatic Computing

Bachelor of Science in Information Technology (BSIT)

- 1. Software Development
 - a. Software Customization
 - b. Information System Development for an actual client
 - c. Web Application Development (with alpha testing on live servers)
 - d. Mobile Computing Systems
 - e. Multimedia Systems
- 2. Network Design and Implementation and Server Farm Configuration and Management
- 3. IT Management
 - a. IT Strategic Plan for sufficiently complex enterprises
 - b. IT Security Analysis Planning and Implementation

Bachelor of Science in Library and Information Science (BSLIS)

- 1. Managing Information Resources
- 2. Managing Information Services
- 3. Managing Information Tools and Technologies
- 4. Managing Information Organization

RESEARCH PROCESS

**Timetable will vary for Architecture, Library and Information Science, and Information Technology*

	ACTIVITIES	RESPONSIBLE PERSON	TEMPLATE	TIMETABLE
1	Selecting a Research Topic			
	a. Identify at least three (3) potential research topics and its corresponding objectives. b. Furnish the Title Proposal Template A c. Submit accomplished template and defend the titles for evaluation of the research teacher. d. Research teacher initially evaluates and approves the topics before the Title Defense. e. There should be at least three (3) approved research topics before the Title Defense. The group may propose another topic in case the teacher rejects one of the topics presented.	Student Student Student Research Teacher	Title Proposal Template A	Research Methods Week 1 to Week 14
2	Title Defense			
	a. Submit the working titles to the research teacher. b. The research teacher will schedule the group for a title defense. c. The panel of experts will confirm and approve one research title. d. The research teacher and panel of experts will assign one (1) thesis/technical adviser for each group. e. Furnish the Appointment of Research Advisor Form and submit to the research teacher upon confirmation.	Student Research Teacher Panel Research Teacher & Panel Student	Title Proposal Template B Title Proposal Score Sheet Appointment of Research Advisor	Research Methods Week 15 to Week 18
3	Research Proposal Writing			

	a. Researchers starts writing Chapter 1 to Chapter 3 of the approved research title. b. The research adviser, from time to time, monitors the progress of the research proposal. c. The researchers prepare a Gantt Chart and submit to the research teacher.	Student Adviser Student	Manuscript Adviser Monitoring Sheet Gantt Chart	Research 1 Week 1 to Week 14
4	Proposal Defense			
	a. The research teacher prepares the schedule of each group for the Proposal Defense b. The researchers furnish the Endorsement and Request for Proposal Defense. c. The adviser evaluates the manuscript and endorses the group for proposal defense. d. The research teacher identifies two (2) panel of experts to sit as research evaluator and critic and one (1) cross-panel. e. Researchers furnish the Appointment of Research Evaluator and submit to research teacher upon confirmation. f. The researchers distribute the four (4) copies of proposal manuscript one (1) week prior to scheduled defense. One (1) copy each for the research adviser and two (2) panel of experts and one (1) cross panel. g. The panel of experts and cross-panel recommend and evaluate the proposal during the defense. h. Follow the comments, evaluation and final recommendation of the panelists. After the revision, proceed to the data gathering procedure, testing and	Research Teacher Student Research Adviser Research Teacher Student Student Panel Student	Endorsement and Request for Proposal Defense Appointment of Research Evaluator Manuscript Proposal Defense Score Sheet and Proposal Evaluation Sheet Revised Manuscript	Research 1 Week 15 to Week 18

	prototyping.			
5	Research Conduct, Writing, Prototyping, Testing & Evaluation			
	a. Researchers starts writing Chapter 4 to Chapter 5 of the approved research title. b. The research adviser, from time to time, monitors the progress of the research. c. Complete the manuscript.	Student Adviser Student	Manuscript Adviser Monitoring Sheet Manuscript	Research 2 Week 1 to Week 10
6	Final Defense			
	a. The research teacher prepares the schedule of each group for the Final Defense b. The researchers furnish the Endorsement and Request for Final Defense. c. The adviser evaluates the manuscript and endorses the group for Final defense. d. The researchers distribute the four (4) copies of final manuscript one (1) week prior to scheduled defense. One (1) copy each for the research adviser and two (2) panel of experts and one (1) cross panel. e. The panel of experts and cross-panel recommend and evaluate the paper during the defense.	Student Student Research Adviser Student Panel	Endorsement and Request for Final Defense. Manuscript Final Defense Score Sheet and Final Defense Evaluation Sheet	Research 2 Week 11 to Week 18
7	Submission of Final Manuscript			
	a. Follow the comments, evaluation and final recommendation of the panelists. b. Revision of the manuscript following the standard format and styles. c. Furnish the Routing Slip Form and follow as advised. d. Research teacher collects the accomplished Routing Slip Form	Student Student Student Research Teacher	Routing Slip Form Final Manuscript	Clearance Period

SELECTING A RESEARCH TOPIC

Topic selection is one of the most difficult tasks in engineering, architecture and technology research paper writing. This is mostly due to the fact that majority of students have more than one favorite topic and so they doubt which one to choose for their final project paper.

Considerations in Selecting a Topic

1. The research to be selected must be aligned in the priority areas of a particular discipline and to the United Nations Sustainable Development Goals. Let us say for BS Civil Engineering;

Topic: Road Safety Assessment of Public Schools in Bacoar City, Cavite

The topic selected is under “Project Management”, which is one of the research priority areas under BS Civil Engineering. This also aligns with the UN SGD 11 “Sustainable Cities and Communities”

2. The choice of topic must be characterized by originality and uniqueness. However, development of a previous research can be considered depending on the level of development that the researcher will going to do.

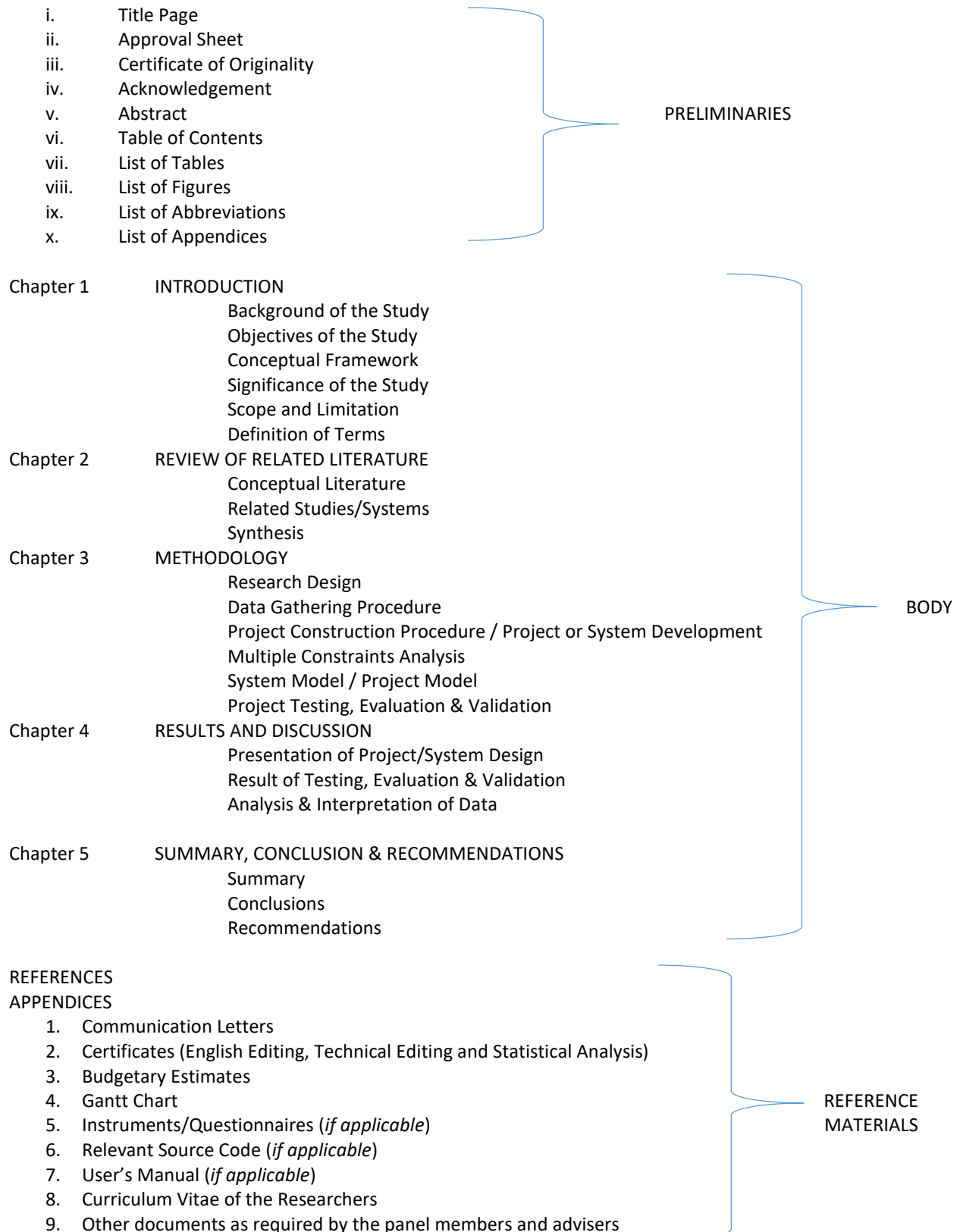
Let us say for example there is an existing research entitled “*Development of a Soldering Iron Device with Temperature Display*”, the control system used was “Bang-Bang” Control. In this case, another research can be developed by using another mode of control, (e.g., PID Control, Proportional Control) to improve the efficiency and overall output of the device.

Most of the time, the researcher recommends or suggests what else can be done to improve their research. It can be found in the last part of the chapter of the final paper under “Recommendation”. The development to be done must be clearly stated and established during the writing of the research proposal.

It is very important to note that the selected topic must have contribution to new knowledge and that there were new insights drawn from it.

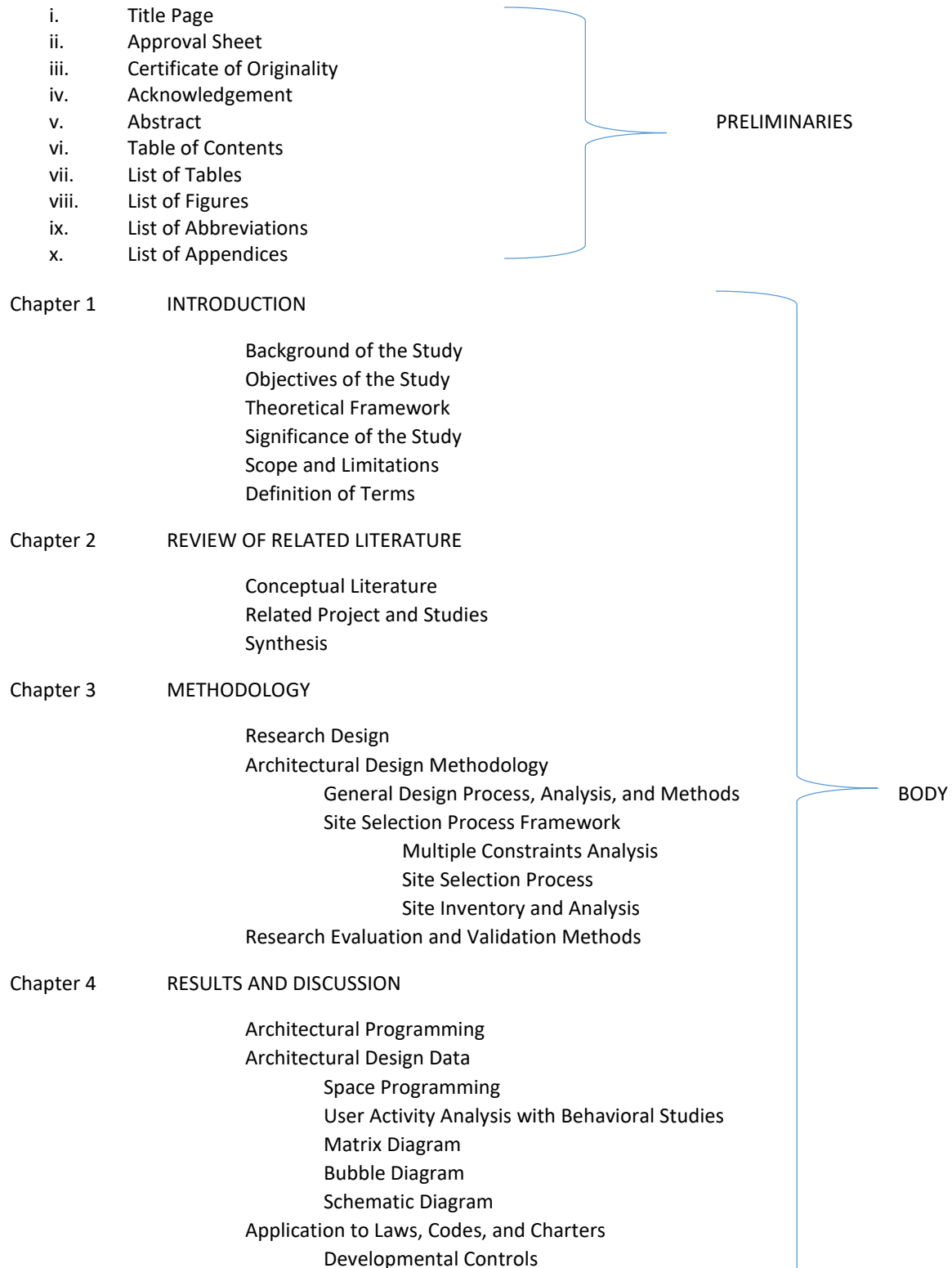
3. The researcher must also consider the significance of the study. Does it have national implications? Will it benefit or impact others in part or whole? Students must not only consider selecting a topic for the benefit of passing the subject but also the benefit the research will contribute to society.
4. An engineering research must present high level of technical quality. Researchers must gain sufficient knowledge as an engineering student. It is an important aspect of the final year project since it is the only official document the students submit together with the developed device if applicable.

MANUSCRIPT OUTLINE FOR ENGINEERING AND ARCHITECTURE RESEARCH



MANUSCRIPT OUTLINE FOR ARCHITECTURAL DESIGN PROJECT

GENERAL RESEARCH CONTENTS (BOOK 1)



	- Parking Requirements - Others - Site Selection, Analysis, and Programming - Site Selection Derivation Framework - Site Inventory - Site Analysis - Macro Site Analysis - Micro Site Analysis - Conceptual Design Framework - Architectural Solution Diagram - Project Goals and Objectives - Design Concepts, Strategies, and Frameworks - Design and Construction Methods - Utilities and Sustainability Methods - Design Development Studies - Design Exploration Studies - Design Parameters - Morphological Development Process - Architectural Solutions, Strategies, and Design Translation	BODY
Chapter 5	SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS - Summary - Conclusions - Recommendations	
	MATRIX OF APPENDICES (BOOK 2) Site Selection Derivation Maps Design Exploration and Process Compilation of Technical Drawings (11" X 17") - Perspectives - Master Development Plan - Site Features and Amenities - Perspective (Aerial and Spot) - Plans - Section - Buildings - Perspectives (Exterior and Interior) - Site Development Plan - Floor Plans - Elevations - Sections - Reflected Ceiling Plan - Architectural Bay Section	REFERENCE MATERIALS

MANUSCRIPT OUTLINE FOR IT, LIS and CS

x.	Title Page		
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	Multiple Constraints Analysis		
	System Development Process		
	System Architecture		
	System Testing, Evaluation and Validation		
	Ethical Considerations		
Chapter 4	RESULTS AND DISCUSSION		
	Presentation of Project/System Design		
	Result of Testing, Evaluation & Validation		
	Analysis & Interpretation of Data		
Chapter 5	SUMMARY, CONCLUSION & RECOMMENDATIONS		
	Summary		
	Conclusions		
	Recommendations		
			BODY
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	11. Certificates (English Editing, Technical Editing and Statistical Analysis)		
	12. Budgetary Estimates		
	13. Gantt Chart		
	14. Instruments/Questionnaires (<i>if applicable</i>)		
	15. Relevant Source Code (<i>if applicable</i>)		
	16. User's Manual (<i>if applicable</i>)		
	17. Curriculum Vitae of the Researchers		
	18. Other documents as required by the panel members and advisers		
			REFERENCE MATERIALS

PRELIMINARIES

1. Title Page

The first page of the manuscript is the Title page. It indicates the title of the study, the submission statement, and the degree earned, the name(s) of the author(s), and the batch month and year of completion ^[1]

Title

The maximum length is 15 words

All in uppercase and in boldface

Font style: Times New Roman; font size: 14

Arranged on top of the page in an inverted pyramid form

Center aligned and Single spaced

Submission Statement

Starts at 2.25" from the left ruler or four-five spaces after the title

Center aligned and Single spaced

Paragraph heading style

First line contains the Nature of the Research/Manuscript

Second line indicates the college or department to which the study is presented

Third line shows the complete name of the Institution

Degree Earned

Starts at 4.25" from the left ruler or four-five spaces after the submission statement

Center aligned and Single spaced

Paragraph heading style

First line: In Partial Fulfilment

Second line: of the Requirement for the Degree

Third line indicates the Degree

Author's Name

Starts at 6.25" from the left ruler or four-five spaces after the Degree Center aligned and boldface

All capital letters

Month and Year of Completion

Written on the bottom part of the page

Center aligned

Paragraph heading style

Without comma

^[1] *Inductivo I., LPU Student Research Manual Undergraduate pp. 96 (2014)*

SAMPLE TITLE PAGE

LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE

**DEVELOPMENT OF A TEMPERATURE CONTROLLED
SOLDERING IRON DEVICE WITH DIGITAL DISPLAY**

A Thesis
Submitted to the Faculty of the
College of Engineering, Computer Studies and Architecture
Lyceum of the Philippines University - Cavite

In Partial Fulfillment
of the Requirements of the Degree
Bachelor of Science in Electrical Engineering

**SHERRY ROSE B. ARNESTO
NANETTE E. BALDOMERO
MARIA CRISELDA G. CALUSIM
LEILANI T. MAGTIBAY
EANNE MAE D. TAMBIGA**

MARCH 2024

COLLEGE OF ENGINEERING, COMPUTER STUDIES AND ARCHITECTURE

2. Approval Sheet

It indicates the formal accomplishment, acceptance and rating of the research. It signifies that the study has been successfully presented or defended. It contains the title of the study, the author(s), the degree, the names and signatures of the Research Adviser who endorsed, Chair and Members of the Panel of Examiners who approved, and the Vice President for Academic Affairs who accepted the paper. ^[1]

Name of College or Department

Written immediately below name of University,
Paragraph heading style and boldface;

Approval Sheet Title

Written three (3) spaces beneath the Name of the College or Department
Written in uppercase and boldface with expanded (2.0) character spacing ^[1]

Research Information Clause

Written in full block paragraph form. It contains the title of the study in uppercase and boldface, the name of the proponent(s) in uppercase and boldface, and the degree sought in paragraph heading style. ^[1]

Adviser's Signature

The name of the adviser should be placed 2-3 spaces beneath the Research Information Clause, right aligned, in uppercase, in boldface, and a designation of Research Adviser (Italics, center aligned) placed under the name. It should be signed by the Adviser. No underscore/underline should be provided. ^[1]

Approval and Acceptance Clause

The approval clause, placed 2 spaces after the adviser, consists of the written approval and acceptance of the committee with the grade of the researcher(s) and the date of the defense. ^[1]

Panel of Examiners

Three (3) spaces underneath the clause are the names of the panelists, the Dean, and the Vice President for Academic Affairs. Names of the signatories should be in uppercase and in boldface The panel of experts are place on each side above the Dean. The name of the Dean must be written at the center The name of the Vice President for Academic Affairs is placed at the bottom page, centered. Designation of signatories is indicated one space below their name in Paragraph Heading style, Italics; No underscore/underline should be provided. ^[1]

^[1] *Inductivo I., LPU Student Research Manual Undergraduate pp. 43 (2014)*

SAMPLE APPROVAL SHEET

	LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE	
	APPROVAL SHEET This thesis entitled DEVELOPMENT TEMPERATURE CONTROLLED SOLDERING IRON DEVICE WITH TEMPERATURE DISPLAY prepared and submitted by SHERRY ROSE B. ARNESTO, NANETTE E. BALDOMERO, MARIA CRISELDA G. CALUSIM, LEILANI T. MAGTIBAY and EANNE MAE D. TAMBIGA has been reviewed and recommended as partial fulfillment of the requirements for the degree of Bachelor of Science in Electronics Engineering ENGR. REX PEÑAS <i>Research Adviser</i> Accepted and approved by the panel of examiners following a successful Oral Defense on March 24, 2024 with a grade of _____. DR. LEAH Q. SANTOS <i>Panel Member</i> ENGR. RHIA V. ESPANTO <i>Panel Member</i> DR. DELIA FAINSAN <i>Panel Chairman</i> Accepted and approved in partial fulfillment of the requirements for the degree of Bachelor of Science in Electronics Engineering DR. ARNEL M. AVELINO <i>Dean</i> DR. MARK IRVIN C. CELIS <i>Vice President for Academic Affairs</i>	
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3. Certificate of Originality

As part of the ethics of the conduct of research and of the institution's sincere commitment to uphold the intellectual proprietary rights, the certificate of originality shall be furnished and be included in the manuscript by all researchers.

The Certificate of Originality is a testament of the researchers that the intellectual content of his/her manuscript is the product of his/her own work and that, to the best of his/her knowledge and belief, it contains no material previously published or written by another person nor material to which to a substantial extent has been accepted for award of any other degree or diploma of a university or other institute of higher learning, except where due acknowledgement is made in the text.^[1]

The first scrutiny of originality is assessed and done during the submission of the working title proposal. Titles, with its covariates, target participants and potential instruments, are verified if such studies have been previously conducted, replicated or modified. No repetition of actual research will be allowed. Replication of study in different participants, subjects or locale are permissible provided that consent was secured from the original proponent(s) of the study. Modified researches with instruments used in a previous study must also secure a consent/permission from the owner of the instrument(s).

^[1] *Inductivo I., LPU Student Research Manual Undergraduate pp. 23 (2014)*

SAMPLE CERTIFICATE OF ORIGINALITY

LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE

LYCEUM OF THE PHILIPPINES UNIVERSITY
Cavite Campus

Certificate of Originality

I/We hereby declare that the research paper entitled:

--

which to a substantial extent has been accepted for award of any other degree or diploma of a university or other institute of higher learning, except where due acknowledgement is made in the text.

I/We also declare that the intellectual content of this thesis is the product of my/our work, even though I/we may have received assistance from others on style, presentation and language expression.

Proponent's Signature

Proponent's Signature

Proponent's Signature

Proponent's Signature

Proponent's Signature

Attested by:

Research Adviser's Signature

Chairperson's Signature

Noted by:

Dr. Arnel M. Avelino
Dean/College Secretary's Signature

Date of Completion: (mm/dd/yyyy)

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4. Acknowledgment

The Acknowledgment page is the section allotted for authors/ researchers/proponents to convey their gratitude to persons who have been beneficial to their study/project.

Title

The title ACKNOWLEDGMENT must be on top of the page, in uppercase, boldface, and center aligned. The institution prefers the American English formulation of acknowledgment, i.e., without the middle e.

Contents

The acknowledgment should be written in the third person; An introductory paragraph is recommended at the beginning; Acknowledge key persons according to professional hierarchy; Filipino and/or foreign words should be italicized; At the end of the acknowledgment, the name of the researcher(s) must be in uppercase and in boldface, must be placed four single spaces after the last sentence, and must be signed properly.

[1]

[1] *Inductivo I., LPU Student Research Manual Undergraduate pp. 23 (2014)*

SAMPLE ACKNOWLEDGEMENT

	LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE	
	ACKNOWLEDGMENT This undergraduate thesis would not have been possible without the support of many people. The researchers wish to express their gratitude to their research teacher and adviser, Prof. Dr. XX who was abundantly helpful and offered invaluable assistance, support and guidance. Deepest gratitude is also due to the members of the panel of examiners, Assoc. Prof. Dr. YY and Dr. ZZ, without their knowledge and assistance this study would not have been successful. Special thanks also to all their graduate friends, especially section members; A, B, C, D and E for sharing their knowledge and invaluable assistance. The researchers would also like to convey thanks to the Faculty for providing the instructions and laboratory facilities. The researchers wish to express their love and gratitude to their beloved family; for their understanding and endless love, through the duration of their studies. Above all, the researchers would like to give their sincerest and warmest appreciation, to the Lord, and Savior, ALMIGHTY GOD, who is the source of their strength, knowledge and skills, who always hears their prayers, guides their path, and never leaves their side throughout their journey in life. APRIL CLIFFORD JOHN NELSON DECENA JOVEN ESCARES ALFRED RAY SABALE	
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5. Abstract

Research abstracts are used throughout the research community to provide a concise description about a research project. It is typically a short summary of your completed research. If done well, it makes the reader want to learn more about your research. Some students present their research findings at local and national conferences. Research abstracts are usually requested as part of the application process for conference presenters. These are the basic components of an abstract in any discipline:

1. Motivation/problem statement: Why do we care about the problem? What practical, scientific, theoretical or artistic gap is your research filling?
2. Methods/procedure/approach: What did you actually do to get your results? (e.g. types of testing and evaluation, prototyping, survey and interviews, equation/formula etc)
3. Results/findings/product: As a result of completing the above procedure, what did you learn/invent/create?
4. Conclusion/implications: What are the larger implications of your findings, especially for the problem/gap identified in step 1?

SAMPLE ABSTRACT

LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE

ABSTRACT

Design and Fabrication of an Allium Cepa Seed-Sowing Machine

Edronne Ron A. Cabanlit¹, Kim Patrick Dimailig², Emanuel Adrian Inandan³, and
Engr. Jessa Ornopia⁴
Proponents¹²³ Adviser⁴

The paper describes the design, development, and fabrication of a machine specifically created to sow Allium cepa seeds, more commonly known as onion seeds. This machine aims to tackle the limitations of manual sowing, such as its inefficiency, inconsistency in both seed spacing and depth, and the high labor requirements it demands. The design prioritizes creating a machine well-suited for small-scale farms, keeping affordability, ease of operation, and the use of readily available materials at the forefront. The paper delves into the selection of components, fabrication processes, and the overall functionality of the machine.

This research contributes significantly to the development of practical tools for farmers by aiming to improve planting efficiency and potentially increase onion yield. By providing a more streamlined and consistent sowing method, this machine has the potential to revolutionize small-scale onion farming. Imagine fields planted with perfect rows of onions, each seed strategically placed for optimal growth. This translates to less wasted seeds, a more uniform crop, and potentially higher yields. Furthermore, the reduced labor requirements associated with the machine could free up valuable time for farmers to focus on other aspects of their

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6. Table of Contents

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a. List of Tables or Matrices (if any)

Title

The title, LIST OF TABLES, should be in uppercase, boldface and centered on top of the page. If there is only one table presented in the entire manuscript the title will be simply “TABLE” instead or the page becomes optional.

Captions The word “Table” is flushed right opposite the word “Page” which is flushed left. Both of them are boldface, found two spaces below the title, LIST OF TABLES;

First Column Table numbers on the first column are in Arabic numerals, consecutively;

Second Column Table titles found immediately after the table numbers are in sentence heading style; Titles must follow proper citation. Table titles are typed double-spaced between entries, except when a title is too long to be typed on one line, it is continued on the next line placing the second and succeeding lines in a 0.5” space hanging indent form.

Third Column The pages where the tables are found are also in Arabic numerals; Table titles and page numbers are connected by light lines (dot for every space) for easy reference; and each line should be one space away from each other.

Sample List of Tables

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Title

The title, LIST OF FIGURES, should be in uppercase, boldface and centered on top of the page. If there is only one table presented in the entire manuscript the page should be omitted.

Captions The word “Figure” is flushed left opposite the word “Page” which is flushed right. Both of them are boldface, found two spaces below the title, LIST OF FIGURES;

First Column Figure numbers on the first column are in Arabic numerals, consecutively;

Second Column Figure titles found immediately after the figure numbers are in sentence heading style; Figures must follow proper citation. Figure titles are typed double-spaced between entries, except when a title is too long to be typed on one line, it is continued on the next line placing the second and succeeding lines in a 0.5” space hanging indent form.

Third Column The pages where the tables are found are also in Arabic numerals; Figure titles and page numbers are connected by light lines (dot for every space) for easy reference; and each line should be one space away from each other.

Sample List of Figure

LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE		
LIST OF FIGURES		
Figure		Page
1	Conceptual framework of the study.....	4
2	Existing food chain in the area (Alvarez, 2009).....	16
3	Lead sources and routes of exposure (ATSDR, 2000).....	17
4	Map of the study area.....	28
5	Profile of the participants according to age.....	32

c. List of Appendices (if any)

The LIST OF APPENDICES is an enumeration of all the information, data and procedures employed in the conduct of the study. Each item should be labelled properly using the same rules in writing figure or table captions. The list may be arranged according to the importance or significance of the material or in alphabetical order.

Title

The title, LIST OF APPENDICES, should be in uppercase, boldface and centered on top of the page. If there is only one table presented in the entire manuscript the page should be omitted.

Captions The word “Appendix” is flushed left opposite the word “Page” which is flushed right. Both of them are boldface, found two spaces below the title, LIST OF APPENDICES;

First Column Appendix numbers on the first column are in Arabic numerals, consecutively;

Second Column Appendix titles found immediately after the figure numbers are in sentence heading style; each appendix must follow proper citation. Appendix titles are typed double-spaced between entries, except when a title is too long to be typed on one line, it is continued on the next line placing the second and succeeding lines in a 0.5” space hanging indent form.

Third Column The pages where the tables are found are also in Arabic numerals; Appendices and page numbers are connected by light lines (dot for every space) for easy reference; and each line should be one space away from each other.

Sample List of Appendices

LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE		
LIST OF APPENDICES		
Appendix		Page
1	Permit to conduct the interviews	54
2	Informed consent for research participants.....	55
3	Survey questionnaires.....	67
4	Documentary analysis.....	68

The Body

CHAPTER 1 Introduction

Chapter 1, with a highly focused review of the literature, and is normally the “prospectus” that a committee approves before the “proposal” to start research is approved. After the prospectus is approved, some of the review of literature may be moved into Chapter 2, which then becomes part of the proposal to do research.

Chapter 1 is the engine that drives the rest of the document, and it must be a complete empirical argument as is found in courts of law. It should be filled with proofs throughout. It is not a creative writing project in a creative writing class; hence, once a word or phrase is established in Chapter 1, use the same word or phrase throughout the research paper.

1. Introduction

State the general field of interest in one or two paragraphs, and end with a sentence that states what study will accomplish. Do not keep the reader waiting to find out the precise subject of the research project.

a. Background of the Study

This section is critically important as it must contain some mention of all the subject matter in the following Chapter 2 Review of the Literature and the methodology in Chapter 3. Key words should abound that will subsequently be used again in Chapter 2. The section is a brief two to four-page summary of the major findings in the field of interest that cites the most current finding in the subject area. A minimum of two to three citations to the literature per paragraph is advisable. The paragraphs must be a summary of unresolved issues, conflicting findings, social concerns, or educational, national, or international issues, and lead to the next section, objectives of the study. The problem is the gap in the knowledge. The focus of the Background of the Problem is where a gap in the knowledge is found in the current body of empirical (research) literature.

b. Objectives of the Study

Arising from the background of the study is this statement of the exact gap in the knowledge discussed in previous paragraphs that reviewed the most current literature found. A gap in the knowledge is the entire reason for the study, so state it specifically and exactly. Use the words “gap in the knowledge.” The objective of the study will contain a definition of the general objective needed for the study, and the specific objectives that will be accomplished. Please make the objectives SMART (Specific, Measurable, Achievable, Relevant, and Time-Bound). In addition, objectives must be in chronological order to avoid confusion.

For Example:

General Objective

The main objective of this study is to design a device that will produce potable water out of waste water using three (3) water treatment processes.

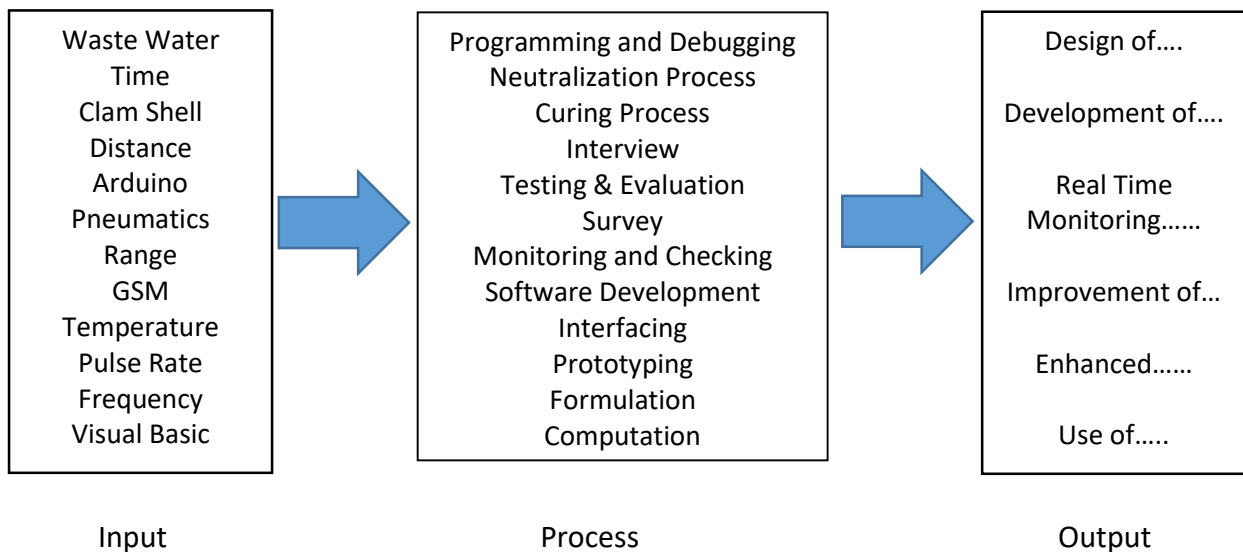
Specific Objectives

Specifically this paper aims to....

- | | |
|----------------------|---------------------------------------|
| 1. To design..... | 11. To interface..... |
| 2. To construct..... | 12. To increase/reduce/lessen/add.... |
| 3. To create..... | 13. To prove.... |
| 4. To develop..... | 14. To replace..... |
| 5. To compare..... | 15. To eliminate..... |
| 6. To improve..... | 16. To measure..... |
| 7. To use..... | 17. To implement.... |
| 8. To test..... | 18. To study..... |
| 9. To evaluate..... | 19. To describe.... |
| 10. To program..... | 20. To relate.... |

c. *Conceptual Framework*

It is the researcher's own position on the problem and gives direction to the study. It may be an adaptation of a model used in a previous study, with modifications to suit the inquiry. Aside from showing the direction of the study, through the conceptual framework, the researcher can be able to show the relationships of the different constructs that he wants to investigate.



d. Significance of the Study

This section discusses the reasons in conducting the research. The aim is quite similar in answering why there is a need to conduct the study. However, in the significance of the study, other questions such as: What benefits would be derived from it? Will the study add to knowledge? Will it have practical application? are also addressed. ^[1]

This section also justifies that it does not duplicate any previous investigation; or if it does, it establishes that this study will have a new contribution to the field. When writing the Significance of the Study, bear in mind the:

- Immediate or long-term benefit accruing to the end user
- Value of the investigation to various individuals, groups or sectors, community, and nation.
- The sectors that will benefit from the study but make sure to specify and explain how it will help them.

e. Scope and Limitation of the Study

Scope refers to the parameters under which the study will be operating. The problem that the researcher seek to resolve will fit within certain parameters. Think of the scope as the domain of the research – what's in the domain, and what is not. It must be as clear as possible what will be studying and what factors are within the accepted range of the study. ^[1]

Limitation are matters, parameters or occurrences which are out of the researcher's control. They limit the extensity to which a study can go, and sometimes affect the end result and conclusion that can be drawn. ^[1]

f. Definition of Terms

Important words, specialized terms or technical expressions should be given their authoritative and/or operational definition in the context of the study. Definitions should have authoritative basis (taken from books, etc.), thus sources are cited. Operational definitions are usually based on authoritative definitions, although with slight variations to suit the needs of the proposed study. ^[1]

^{1]} *Inductivo I., LPU Student Research Manual Undergraduate pp. 53 (2014)*

CHAPTER 2

Review of Related Literature

The purpose of a review is to analyze critically a segment of a published body of knowledge through summary, classification, and comparison of prior research studies, reviews of literature, and theoretical articles. A literature review is an account of what has been published “related” to the topic of interest. It includes summary of studies, inquiries, or investigations conducted for the last five to ten years, except for primary sources. It may indicate a gap or continuity between the previous and the present studies. The background of earlier studies relating to the problem at hand should be developed to show that the existing evidence does not solve the problem adequately; hence, the need to provide new insights and more relevant information, methods and solutions. ^[1]

Importance of Literature Review

- Serves as a source of research problems which can be gathered from existing studies.
- Ensures that important variables relevant to the study are not overlooked.
- Find gaps in the literature. ☐ Learn about opposing views.
- Establishes the need /significance / justification in conducting the present study.
- Serves as a basis for the development of the theoretical / conceptual framework.
- Avoids the unnecessary duplication of selecting a research problem which has been done before.
- Helps give more focus, direction and clarity to the study.
- Learns about other people working in the same field (a comparative and cumulative list of the development of the current study).
- Provides the intellectual context for the project.
- Identifies research methods that could be relevant to the project.

Conceptual Literature

Conceptual literature reviews focus on theoretical frameworks, conceptual models, and foundational theories relevant to the research topic. The purpose of a conceptual literature review is to provide a theoretical foundation for the study, explaining key concepts, defining terms, and outlining theoretical frameworks that inform the research.

Conceptual literature reviews are ideas connected to your current study that can be found in books, encyclopedias, government or institutional reports, monographs, professional association reports, and other materials where frameworks, laws and foundational theories can be found.

Related Studies

A review of related studies focuses on empirical research studies, experiments, surveys, case studies, and other empirical investigations relevant to the research topic. The purpose of a review of related studies is to summarize and evaluate existing empirical research findings, methodologies, and outcomes related to the research question or objectives.

On the other hand, related studies are articles or results that can be found in journals, reports and other sources where empirical or experimental results are presented that can help your current study in proving theories or methods to be utilized for the completion of the study.

Synthesis

After reviewing series of related literature, the synthesis part provides a synopsis of the relevant highlights, similarities, differences and gaps of previous studies. Then, it should provide the critique and rationale why the current study must be undertaken. A synthesis must be brief and concise with two to three paragraphs ^[1]

^[1] *Inductivo I., LPU Student Research Manual Undergraduate pp. 54 (2014)*

CHAPTER 3

Methodology

The Method section describes in detail HOW the study was conducted. A well laid out and logical methodology will provide a great backbone for the entire research paper, and will allow you to build an extremely strong results.

1. Research Design

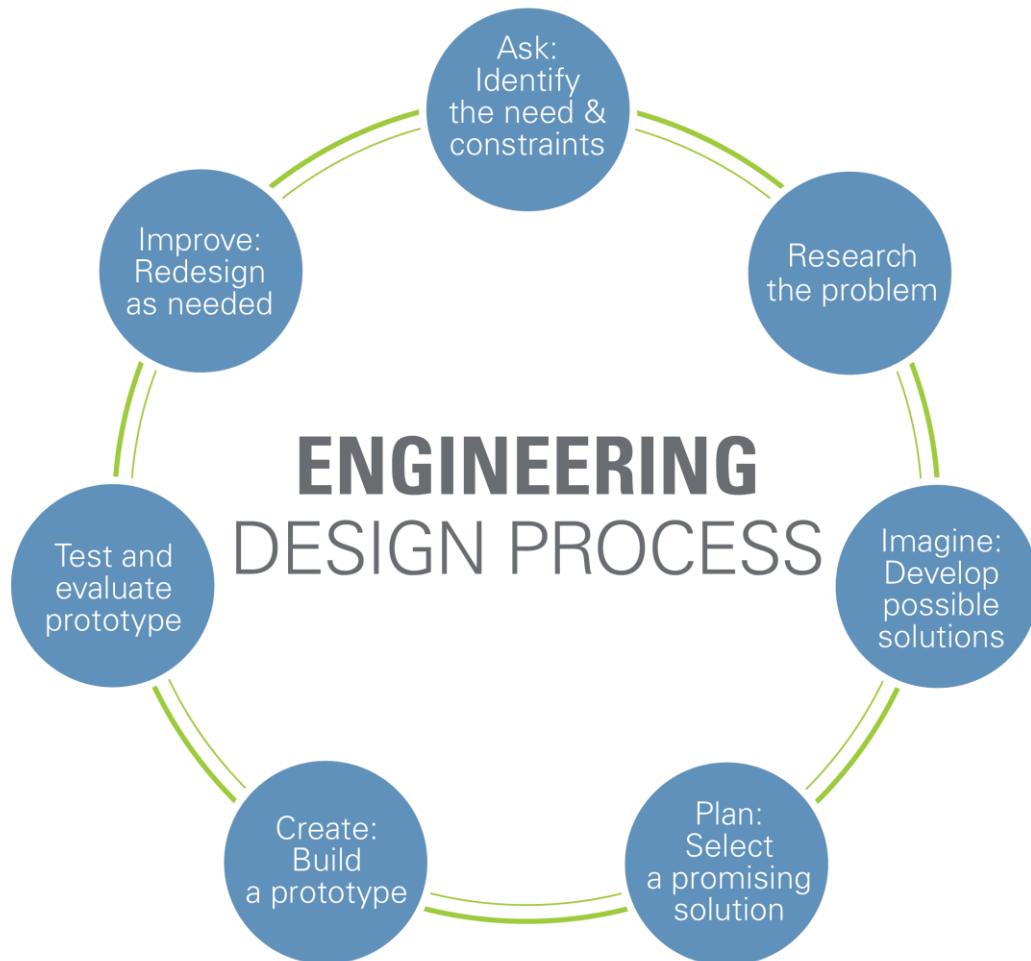
- The **engineering, architecture and technology design process** is a series of steps that engineers follow to come up with a solution to a problem. Many times the solution involves designing a product (like a machine or computer code) that meets certain criteria and/or accomplishes a certain task.
- This process is different from the Steps of the **Scientific Method**. If the project involves making observations and doing experiments, follow the Scientific Method. If the project involves designing, building, and testing something, that should probably follow the Engineering Design Process.
- The steps of the engineering design process are to:
 - a) Define the Problem
 - b) Do Background Research
 - c) Specify Requirements
 - d) Brainstorm Solutions
 - e) Choose the Best Solution
 - f) Do Development Work
 - g) Build a Prototype/Specimen/System Model/Software
 - h) Test and Redesign
- Engineers do not always follow the engineering design process steps in order, one after another. It is very common to design something, test it, find a problem, and then go back to an earlier step to make a modification or change to your design. This way of working is called **iteration**, and it is likely that the process will do the same!

2. Data Gathering

Background research is especially important for engineering design projects, because you can learn from the experience of others rather than blunder around and repeat their mistakes. To make a background research plan— a roadmap of the research questions you need to answer -- follow these steps:

- Identify questions to ask about the target **user** or customer.
- Identify questions to ask about the products that already exist to solve the problem being defined or a problem that is very similar.

- Plan to research how the product will work and how to make it.



Steps to Engineering Design

3. Project Construction Procedure / Project or System Development

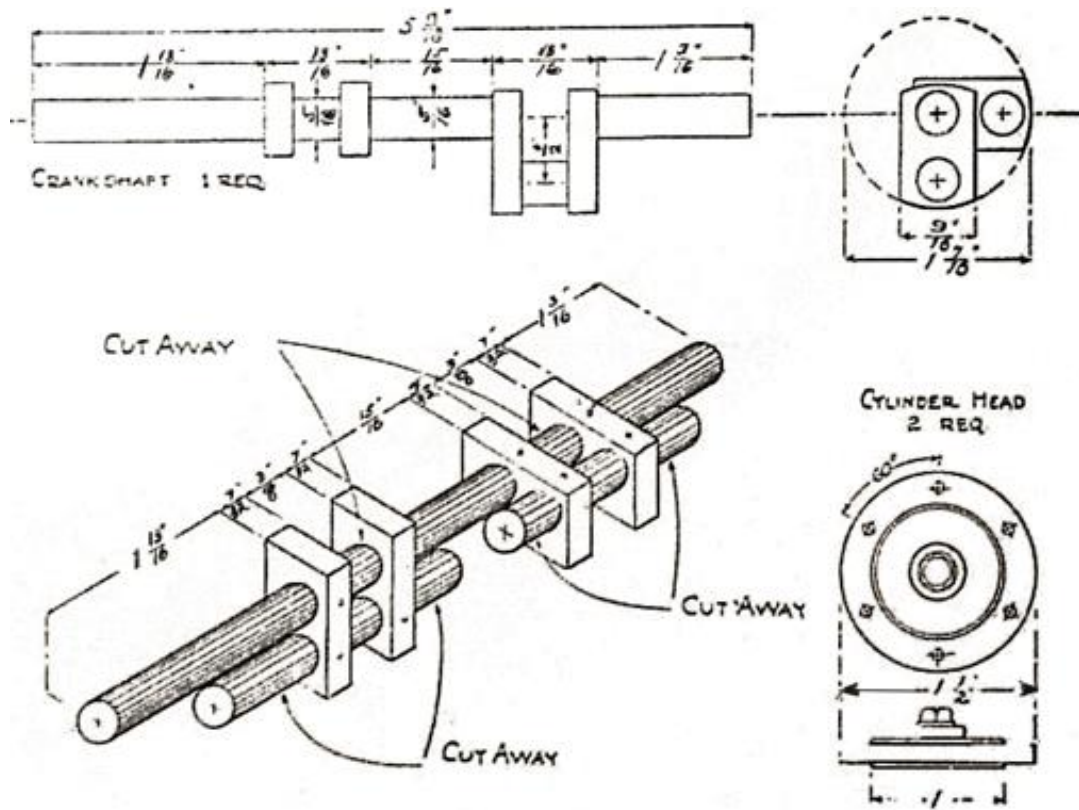
This section will discuss the step by step procedure of how the project will be developed. It will serve as guide of the researchers in constructing the project as a whole. Sample procedures can be seen in Figures Sample A to E

- a) **Design requirements** state the important characteristics that the design must meet in order to be successful.

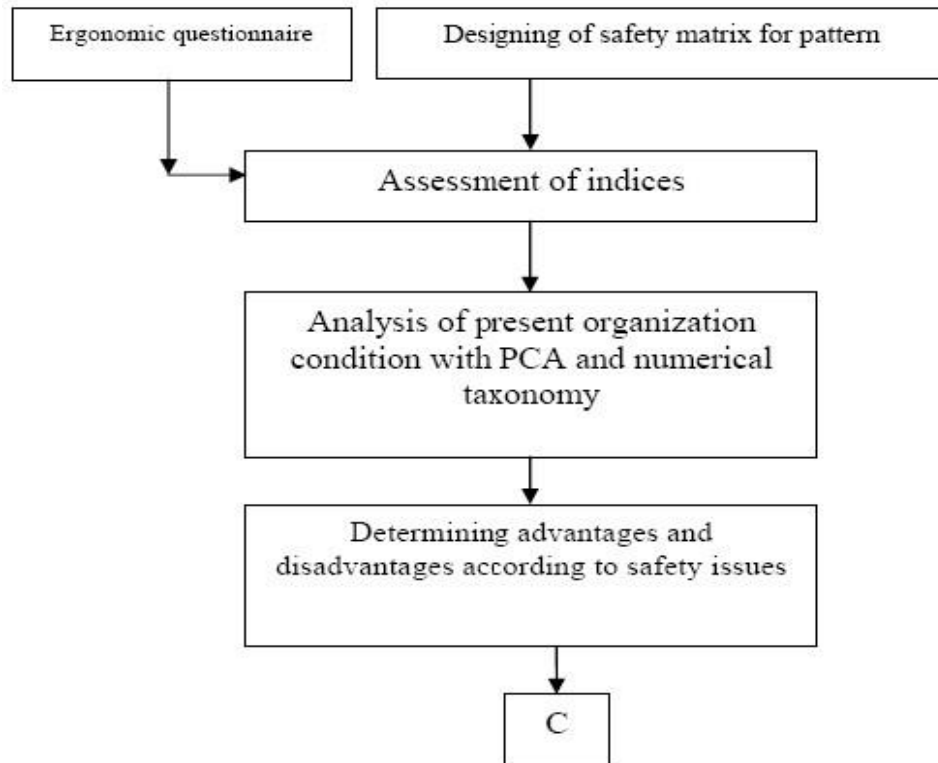
- b) At this point, the researcher must identify the specific requirements that will be used in the project *construction (software needed, programming language, IC, mathematical equation/theory, technical specifications of the materials etc.)*
- c) One of the best ways to identify the design requirements for your solution is to use the concrete example of a similar, existing product, noting each of its key features.

4. System Model / Project Model

Model includes drawing or a block diagram of the project.



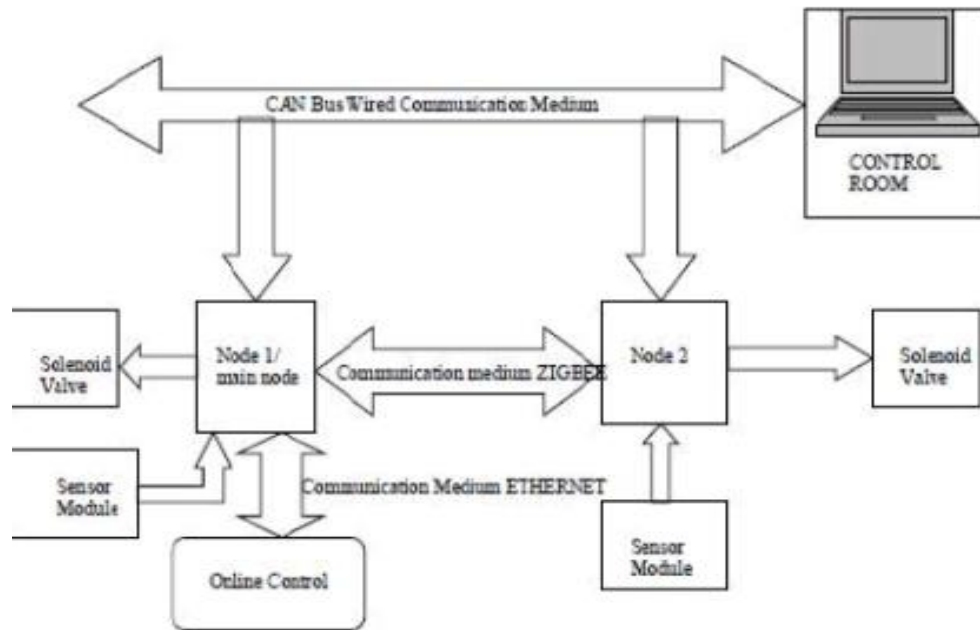
Sample Project Model



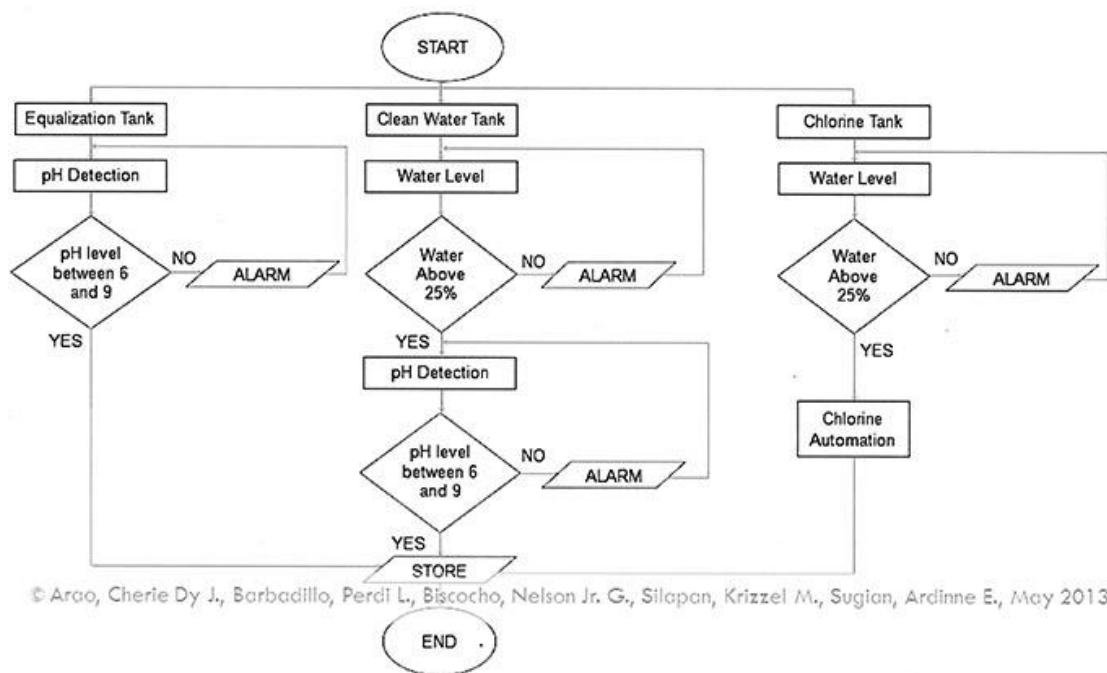
Sample A



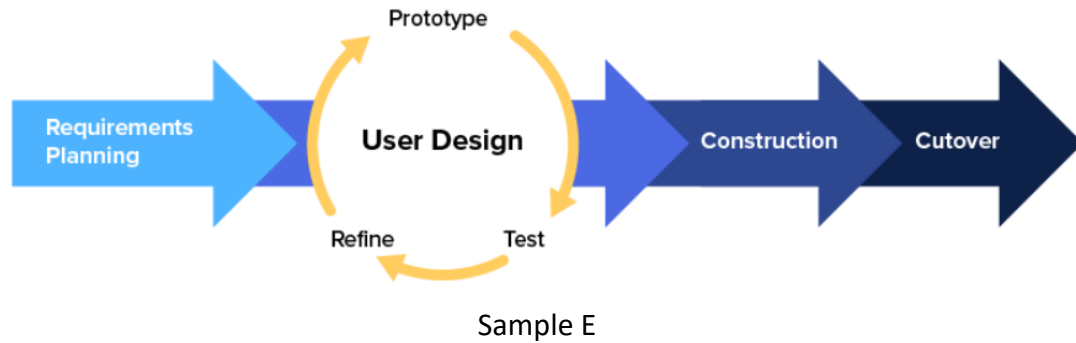
Sample B



Sample C



Sample D



5. Project Testing, Evaluation and Validation

Test and evaluation (T&E) provide the means to ensure that a product performs in the intended way and with the intended results. Testing starts very early in a product system development, however, because test planning is dependent on good requirements.

Test Requirements – Requirements as the primary method for measurement and control of systems development. Where requirements come from; how to evaluate requirements for testability; how to derive test requirements from higher level requirements. Quantifying an operational need; analyzing missions and environments; defining performance, interface, and constraint requirements; evaluation of requirements for testability; deriving test requirements; the importance of a requirements verification matrix (RVM); Qualification vs. Acceptance requirements; design proof vs. first article vs. production requirements.

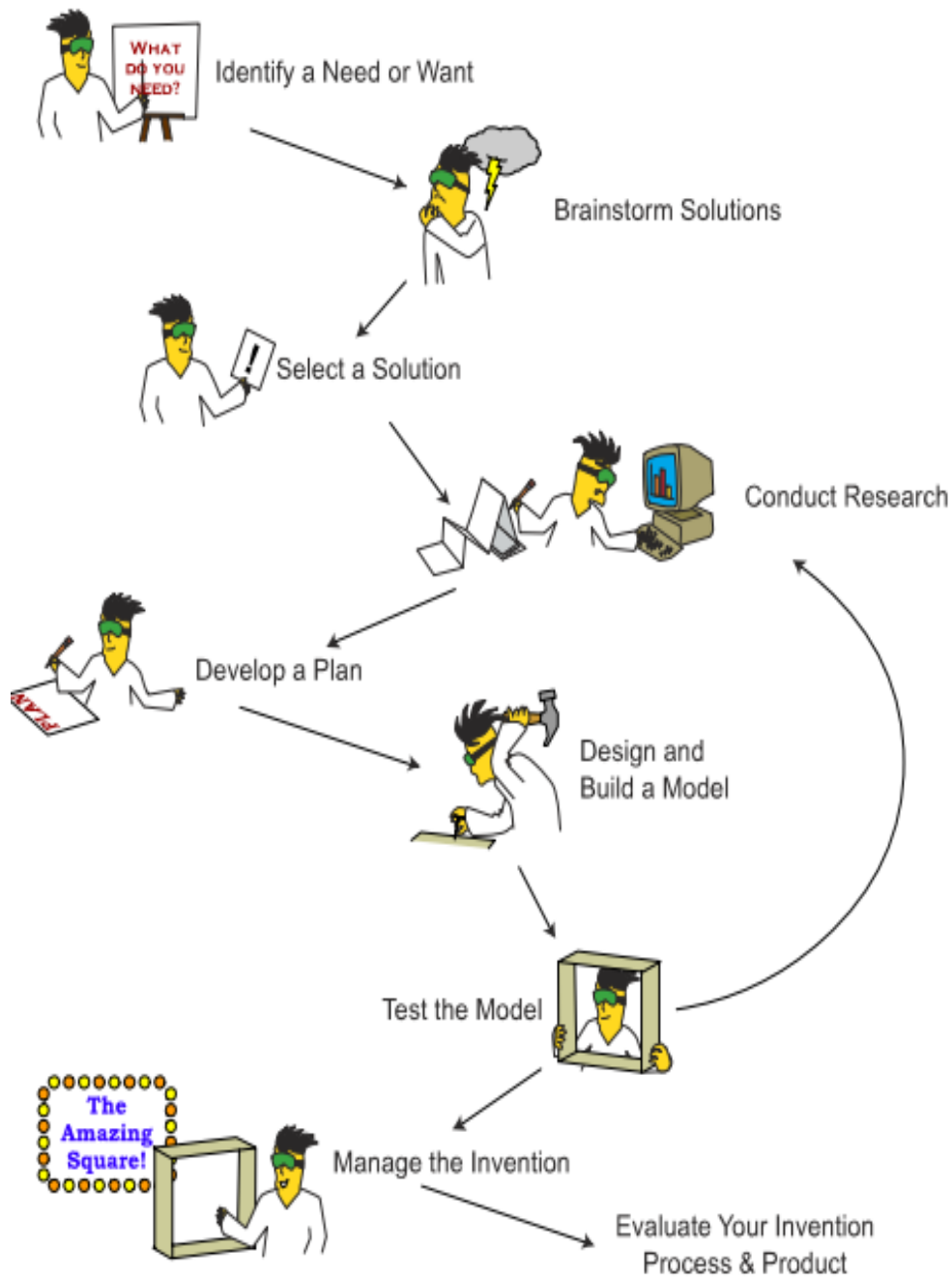
Test Planning – Evaluating the system concept to plan verification and validation by test. Trade-offs involved in test decisions; maturity at different phases; level of detail. System analysis for test planning; analyzing and evaluating alternatives; test resource planning; establishing a test baseline; developing a test schedule; change management; Test and Evaluation Master Plan (TEMP); risk management as a test planning tool.

Integration Testing – How to successfully manage the intricate aspects of system integration testing; level of integration planning; managing complex system integration; work-around. Development test concepts; integration test planning (parallel/serial test sequencing, scheduling); preferred order of events; component testing; conducting integration tests for complex systems; workarounds for anomalies and failures.

Test Conduct – How to perform testing; differences in testing for design proof, first article qualification, recurring production acceptance; rules for test conduct. Testing for different purposes, verification vs. validation; test procedures and test records; prerequisites management; readiness certification, test constraints, test article

configuration; troubleshooting and anomaly handling; measures of success and indicators of difficulty; test tools.

Data Collection, Analysis and Reporting – Statistical methods; test data collection and analysis; report formats and records



The Engineering and Technology Design Process

6. Multiple Constraints Analysis

Multiple constraints analysis is a valuable approach in research, particularly when your study involves complex systems or processes with various factors influencing the outcome. This analysis helps identify, understand, and prioritize constraints or limitations that may affect the results or the overall feasibility of the research.

Start by defining what constraints mean in the context of your research. Constraints can be any factor that limits or restricts the scope, process, or outcomes of your study. The constraints to be analyzed are the following:

- Economical
- Social
- Sustainability
- Environmental
- Health and Safety

Evaluate how each constraint affects the different aspects of your system, design, or methodology. Some constraints may have a minor impact, while others could significantly influence the outcome.

Use techniques such as impact analysis, decision matrices, or sensitivity analysis to prioritize constraints. Focus on those that have the most significant impact on your research objectives and outcomes.

After analyzing the results, consider any additional factors or constraints that might not have been fully captured in the matrix. For example, external factors like potential future developments in the field, or personal expertise in one method over another, might influence your decision.

Example Multiple Constraints Analysis Flow:

You can set up a decision matrix where each row represents one of the methods, and each column represents a constraint. This allows you to systematically compare each method against the same set of criteria. Assign weights to each constraint based on its importance to your research. For example, if economical constraint is more critical than social constraint, you would give it a higher weight.

Develop a scoring system (e.g., 1-5 or 1-10) to rate how well each method meets each constraint. Score each method based on how well it satisfies each constraint. Consider both qualitative and quantitative assessments. Multiply the score for each constraint by its weight to reflect its importance in your decision.

Conduct a sensitivity analysis by adjusting the weights or scores to see how changes in the importance of certain constraints might affect the overall ranking. Determine whether the selected method remains the best choice under different scenarios. This helps ensure that your decision is robust and not overly dependent on any one constraint.

7. Ethical Considerations

Writing ethical considerations for a research thesis manuscript is crucial for demonstrating that your study adheres to ethical standards and respects the rights and well-being of participants and others affected by your research. Ethical considerations should be thoroughly discussed and integrated into various sections of your thesis, particularly in the methodology section.

Start by explaining the importance of ethics in research. Highlight that ethical research protects participants, ensures the integrity of the research process, and maintains public trust in science. Briefly mention why ethical considerations are particularly relevant to your research topic or field.

If applicable, explain how you ensured that participation in your study was voluntary. Participants should have been informed about the nature of the research, what it involved, and their right to withdraw at any time without penalty. Describe how you provided clear and understandable information about the study's purpose, procedures, risks, and benefits. This could include providing a detailed consent form. Mention how consent was documented, such as through signed consent forms or recorded verbal consent.

Discuss how you protected participants' personal information. This could involve anonymizing data, using codes instead of names, or storing data securely. Explain the steps taken to ensure that participants' identities and responses were kept confidential. This could include limiting access to data and ensuring that any published data cannot be traced back to individual participants. If applicable, clarify how anonymity was maintained, particularly if sensitive topics were involved.

CHAPTER 4

Results and Discussion

1. Presentation of Project / System Design

This chapter covers the discussion of how the prototype operates based on its intended function, the design computations, features, structures and actual dimension, technical specification and etc. that will provide enough information about the actual project design

Example 1

Based on the program flow chart shown in figure 3.3 & the actual circuit design, here is the sequential order of how the device works. The sequence of the device process is programmed through PIC using the assembly language. Figure 4.1 shows the actual program loading.

- 1. As the device is turned on, the PIC will read the memory for stored temperature & display the last configured setting.*
- 2. The program will determine if either UP/DOWN button was activated.*
- 3. If UP button is activated, the program will increase the target temperature while less than 300, otherwise, if DOWN button is activated, the program will decrease the target temperature while more than 100.*
- 4. The target temperature will be stored in the memory & will display its value.*
- 5. If the real time temperature is greater than the target temperature, it will automatically turn OFF the heater & display the read temperature value.*

If the real time temperature is less than the target temperature, it will automatically turn ON the heater and display the READ temperature value

Example 2

The wall panel of this invention includes a pair of opposed metal facing sheets, each having different side joint configurations. One of the panel sheets is identified as a liner sheet and is the surface which is secured to a supporting framework. The other panel sheet is a facing sheet which is exposed to view and exposed to the weather in external walls. The facing sheet overlies and is generally parallel to the liner sheet. The facing

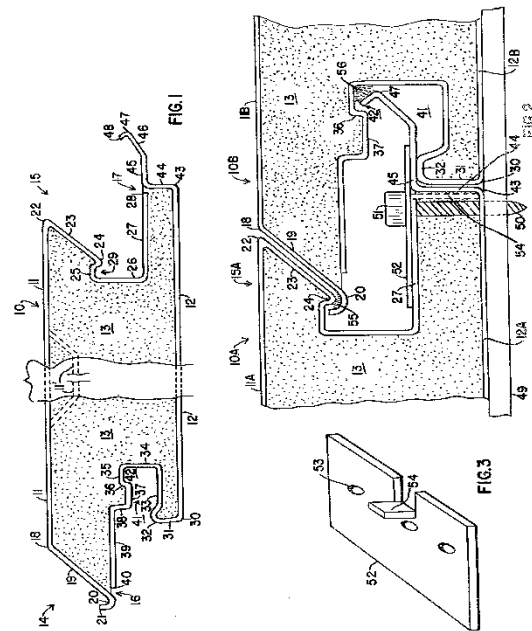
sheet and liner sheet are spaced apart, i.e., they are not in metal-to-metal contact at any point, thereby maintaining outstanding thermal insulating properties for the panel. The liner sheets of abutting panels have wall surfaces extending normally from the building framework which are abutted to produce a wall assembly. The facing sheets of abutting panels have sloping wall surfaces which are in sliding

engagement. For the purposes of the following discussion, two abutting panels will be identified as a first panel and a second panel wherein the first panel is the one which is initially secured to the building framework.

U.S. Patent

Feb. 23, 1982

4,316,351



Example 3

U.S. Patent

Aug. 29, 1996

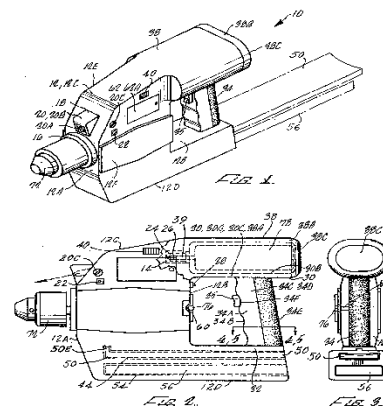
Sheet 1 of 7

5,445,479

As mentioned above, the hand, wrist and forearm support is disclosed and exemplified in three design configuration:

a) in the first design, the support can be retracted into the drill housing when not in use;

b) in the second design, the support is inserted into a support cavity when in use and is removed and stored when not in use; and



c) in the third design, the support is swivel-attached to allow the support to be extended when in use and nested and locked against the lower side of the drill housing when not in use.

2. Result of Testing and Evaluation

In the results section, the summary of the collected data and the analysis performed on those data relevant on the discourse that is to follow are presented. Report the data in sufficient detail to justify the conclusions. Mention all relevant results, including those that run counter to expectation.

Example 1

1) Accuracy Testing

The results shown in table 1 are the data gathered during the testing. Five (5) samples were used to test the accuracy of the prototype that was developed. The group used pH 7.0 and pH 4.0 buffer solutions as two (2) of the samples for the reason that the two were used as the calibration points of the sensor, as recommended on the data sheet of the product. Vinegar and ammonia were used as the extremities of test for acidic and alkaline solution respectively due to the fact that the mentioned two are the most common household solutions that are available to be used. Lastly, the water with fish feed was used as the fifth solution to simulate the actual effect of fish feeds to the pH level of the water when not consumed by the fishes.

TABLE 1. Results for Accuracy Test of pH Sensor

Trial	Vinegar (pH 2.50)	Buffer Solution (pH 4.00)	Water with Fish Feeds (pH 5.00)	Buffer Solution (pH 7.00)	Ammonia (pH 9.60)
1	2.58	4.03	5.19	6.99	9.53
2	2.60	4.01	5.21	6.97	9.53
3	2.61	4.05	5.23	7.00	9.52
4	2.60	4.05	5.24	7.00	9.55
5	2.61	4.06	5.21	7.00	9.57

that was consumed by the system was measured in order to compute the expected battery life. As seen in Table 1 and 2, there are five trials in getting the total current.

Table 3. Measured Total Current Consumed by the System

TRIAL	I (mA)
1	214.1
2	206.2
3	239.2
4	215.0
5	216.3

In order to get the total current, the average of the measured values was computed having the result of 218.6 mA. After getting the average of the measured values, the expected battery life was then computed as shown in the next page.

Example 2

Table 2. Summary of the results of physical and weathering tests.

Sample/test	SMA-2	SMA-6	SMA-8
Specific Gravity (kg/m ³)	2673	2932	2956
Water absorption (%)	1.112	1.194	0.908
% of voids (compacted)	43.27	46.66	47.91
% of voids (loose)	51.27	51.72	54.43
Uniaxial compressive strength (MPa)	59.45	35.22	59.01
Flexural strength (Mpa)	16.53	7.52	8.74
MBA (g/100g)	low OM	low OM	low OM
Los Angeles abrasion (%)	37.82	61.36	46.9
Weathering index	4.75	2.18	0.11
Resistance to desintegration by anhydride sulph	4.75	2.18	0.11

OM, organic matter

Example 3

Comfort descriptors							
This hand tool:	Totally disagree	•	Disagree somewhat	•	Agree somewhat	•	Totally agree
Fits the hand	1	2	3	4	5	6	7
Is functional	1	2	3	4	5	6	7
is easy in use	1	2	3	4	5	6	7
Has a good force transmission	1	2	3	4	5	6	7
Is a high quality tool	1	2	3	4	5	6	7
Has a nice-feeling handle	1	2	3	4	5	6	7
Offers a high task performance	1	2	3	4	5	6	7
Provides a high product quality	1	2	3	4	5	6	7
Looks professional	1	2	3	4	5	6	7
Needs low hand grip force supply	1	2	3	4	5	6	7
Has a good friction between handle and hand	1	2	3	4	5	6	7
Causes an inflamed skin of hand	1	2	3	4	5	6	7
Causes peak pressure on the hand	1	2	3	4	5	6	7
Causes blisters	1	2	3	4	5	6	7
Feels clammy	1	2	3	4	5	6	7
Causes numbness and lack of tactile feeling in hand	1	2	3	4	5	6	7
Causes cramped muscles	1	2	3	4	5	6	7

3. Analysis and Interpretation of Data

The researcher will examine, interpret, and qualify the results and draw inferences and conclusions from the result. The researcher must emphasize any theoretical or practical consequences of the results. The interpretation of the results should take into account (a) sources of potential bias and other threats to internal validity, (b) the imprecision of measures, (c) the overall number of tests or overlap among tests, (d) the effect sizes observed, and (e) other limitations or weaknesses of the study. End the Discussion section with a reasoned and justifiable commentary on the importance of the findings. This concluding section may be brief or extensive provided that it is tightly reasoned, self-contained, and not overstated ^[1]

^[1] *Inductivo I., LPU Student Research Manual Undergraduate pp. 64 (2014)*

Chapter 5

SUMMARY, CONCLUSION, AND RECOMMENDATIONS

This chapter contains the important portions of the study in form of summary. It presents the overview of the study as a whole. The summary, conclusion and recommendations should also cover significant findings in the result during the testing and evaluation.

1. SUMMARY

This portion is a brief presentation of the contents of the thesis. The objectives of the study, methodology and results must be presented in this section. The summary of the findings during the conduct of testing and evaluation should be stated and summarized. The answers to the objective must be relevant to the problems stated in the first chapter of the study.

2. CONCLUSIONS

The conclusions relate directly to the research questions or objectives. They represent the *contribution to the knowledge*. They also relate directly to the significance of the study, which is always, in some way, to improve existing system. These are the major generalizations, the answer to the problem(s) revealed in Chapters 1 and 2. It is in this section that the highlights of the research should be presented. There is no need to discuss the details of the findings but only significant findings should be included. The conclusion should be drawn directly from the objectives of the study and must be answered in an objective and logical manner.

3. RECOMMENDATIONS

These can take two forms: recommendations for further study, or recommendations for change, or both. Each recommendation should trace directly to a conclusion. The researcher may recommend improvement for further study. During the course of the project or research, the researcher might have encountered problems which was not addressed in the study and instead include in a form of recommendation for future study. The researcher may also discuss possible applications of the findings to pertinent systems or situations, and recommend for further study other side problems uncovered during the course of the thesis research.

REFERENCES

Commission on Higher Education [CHED] Memorandum # 53. Series of 2006. Policies and Standards for Information Technology Education (ITE) Programs

Inductivo (2014), LPU Research Manual – Undergraduate , Lyceum of the Philippines University Research Publication and Innovation Center

APPENDIX I

Defense Checklist

CHECKLIST FOR DEFENSE

1. Title Defense Checklist
 - a. Approved Title Proposal Template A
 - b. Title Proposal Template B
 - c. Title Defense Score Sheet
 - d. PowerPoint Presentation
 - e. Contents of the Title Proposal Visual Presentation
 - i. Proposed Title
 - ii. Gap in knowledge
 - iii. Proposed objectives
2. Proposal Defense Checklist
 - a. Endorsement and Request for Proposal Defense
 - b. Three (3) Copies of Printed Soft-Bound Manuscript
 - c. Three (3) Copies of Proposal Score Sheet
 - d. Three (3) Copies of Proposal Evaluation Sheet
 - e. PowerPoint Presentation
 - f. Timetable for Research (Gantt Chart)
 - g. Contents of the (Research Proposal) Visual Presentation
 - i. Title page
 - ii. Brief introduction
 - iii. Objectives of the Study
 - iv. Conceptual Framework
 - v. Significance of the study
 - vi. Scope and Limitation
 - vii. Synthesis of the Literature review
 - viii. Methodology
3. Final Defense Checklist
 - a. Endorsement and Request for Final Defense
 - b. Three (3) Copies of Printed Soft-Bound Manuscript
 - c. Three (3) Copies of Final Defense Score Sheet
 - d. Three (3) Copies of Final Defense Evaluation Sheet
 - e. PowerPoint
 - f. Prototype/Drawing board/video documentation, if applicable
 - g. Contents of the (Research Final Defense) Visual Presentation
 - i. Title page
 - ii. Brief introduction
 - iii. Objectives of the Study
 - iv. Results and Discussions per objective
 - v. Summary
 - vi. Conclusions
 - vii. Recommendations

APPENDIX II

RESEARCH TEMPLATES

List of Research Templates

1. Requisitions
 - a. Title Proposal Template A
 - b. Title Proposal Template B
 - c. Endorsement and Request for Proposal Defense
 - d. Endorsement and Request for Final Defense
2. Evaluation Sheet
 - a. Title Proposal Score Sheet
 - b. Proposal Defense Score Sheet
 - c. Proposal Evaluation Sheet
 - d. Final Defense Score Sheet
 - e. Final Defense Evaluation Sheet
3. Certificates
 - a. Certificate of Originality
 - b. Certificate of English Editing
 - c. Certificate of Technical Editing
 - d. Certificate of Statistical Analysis
4. Reference
 - a. Routing Slip for Final Manuscript
 - b. Adviser Monitoring Sheet
5. Appointments
 - a. Appointment of Research Advisor
 - b. Appointment of Research Evaluator
 - c. Assignment of English Critic
 - d. Assignment of Technical Critic
 - e. Assignment of Statistician

LYCEUM OF THE PHILIPPINES UNIVERSITY
Cavite Campus

Title Proposal Template A

Proponent(s):

A. _____

B. _____

C. _____

D. _____

Proposed Title	Title of Existing Research Projects Connected to the Proposed Title	Proposed Innovation or Development	Remarks A / R
			Research Teacher
			Research Teacher

** A = Accepted * R = Rejected*

**Use extra sheet if necessary*

LYCEUM OF THE PHILIPPINES UNIVERSITY
Cavite Campus

Title Proposal Template B

Proponent(s):

D. _____
E. _____
F. _____
G. _____
H. _____

Degree:

College:

Name of Research Teacher:

___ Sem. A.Y. 20__ - 20__

Working Titles:

1.

Objectives of this study:

2.

Objectives of this study:

3.

Objectives of this study:

**Attach accomplished Title Proposal Template A.*

Comments and Recommendations:

Approved Working Title:

Recommending Approval:

Chairperson's Signature over Printed Name

Date

Approved by:

Dr. Arnel M. Avelino

Dean's Signature

Date

LYCEUM OF THE PHILIPPINES UNIVERSITY
Cavite Campus

Endorsement and Request for Proposal Defense

This is to inform that the research proposal entitled _____
submitted by:

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

with the degree _____ under the College of _____
has/have duly accomplished the requirements needed for proposal defense.

I have reviewed and fully endorse the proposal manuscript attached herewith for oral
review/defense on _____ from _____ to _____ at the _____.
*mm/dd/yyyy**start time**end time**venue*

Research Adviser

We have received copies of the proposal and accept it for oral review/defense on the
aforestated date and venue.

Panelist

Date

Panelist

Date

Panelist

Date

Panelist

Date

Recommending Approval:

Approved by:

Chairperson's Signature over Printed Name

Dr. Arnel M. Avelino
Dean's Signature

Date Completed: _____

Attachment: **Official Receipt of Paid Research Fee**

LYCEUM OF THE PHILIPPINES UNIVERSITY
Cavite Campus

Endorsement and Request for Final Defense

This is to inform that the research manuscript entitled _____

submitted by:

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

with the degree _____ under the College of _____
has/have duly accomplished the requirements needed for final defense.

I have reviewed and fully endorse the final manuscript attached herewith for oral
review/defense on _____ from _____ to _____ at the _____.
mm/dd/yyyy start time end time venue

Research Adviser

We have received copies of the manuscript and accept it for final defense on the foretasted date
and venue.

Panelist

Date

Panelist

Date

Panelist

Date

Panelist

Date

Recommending Approval:

Approved by:

Chairperson's Signature over Printed Name

Dr. Arnel M. Avelino
Dean's Signature

Date Completed: _____

Attachment: **Official Receipt of Paid Research Fee**

LYCEUM OF THE PHILIPPINES UNIVERSITY – CAVITE
College of Engineering, Computer Studies & Architecture

Appointment of Research Advisor

_____, 20__

Mr./Ms./Arch./Engr./Dr. _____
Lyceum of the Philippines University – Cavite
Gen. Trias City, Cavite

Dear Mr./Ms./Arch./Engr./Dr. _____,

Greetings in *Veritas et Fortitudo, Pro Deo et Patria!*

I am pleased to inform you that you have been appointed as Research Advisor of the research study/project entitled: _____

with _____
as your advisee(s).

For your guidance, find attached list of duties and responsibilities.

Sincerely,

Chairpersons Signature

Noted:

Dr. Arnel M. Avelino
Dean's Signature over name

Accepted:

Research Advisor's Signature over Encoded Name

LYCEUM OF THE PHILIPPINES UNIVERSITY – CAVITE
College of Engineering, Computer Studies & Architecture

Appointment of Research Evaluator

_____, 20__

Mr./Ms./Arch./Engr./Dr. _____
Lyceum of the Philippines University – Cavite
Gen. Trias City, Cavite

Dear Mr./Ms./Arch./Engr./Dr. _____,

Greetings in *Veritas et Fortitudo, Pro Deo et Patria!*

I am pleased to inform you that you have been appointed as one of the Research Panel of Examiners/Evaluators of the research study/project entitled: _____

with _____
as your advisee(s).

As an evaluator, the list of appertaining duties and responsibilities are the following:

- a. To critique and evaluate the abovementioned research study according to the standards of research writing, reporting, syntax and format, and with appropriate use of methodologies from title proposal up to final manuscript.
- b. To ascertain that the selection and conduct of the research is relevant, original/innovative, based on the thrusts of the program and exhibits high standards of scholarship.
- c. To stand during defenses as member of the research committee and part of the panel of examiners.

Your expert participation is an adamant sign of your commitment to assure the quality of our students' research output. With gratitude and best regard.

Sincerely,

Chairpersons Signature

Noted:

Dr. Arnel M. Avelino
Dean's Signature over name

Accepted:

Research Evaluators Signature over Encoded Name

LYCEUM OF THE PHILIPPINES UNIVERSITY
Cavite Campus

Assignment of English Critic

This is to inform that the research manuscript entitled _____

submitted by:

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

with the degree _____ under the College of _____
has/have been duly assigned under your faculty for proofreading and English editing with respect to the
standards of proper grammar and composition.

Conforme:

English Critic's Signature over Printed Name

Date Signed: _____
(mm/dd/yyyy)

Approved by:

English Coordinator's Signature over Printed Name

Date Signed: _____
(mm/dd/yyyy)

Note: Reproduce and distribute two (2) duly signed copies to English Critic and Language Coordinator

LYCEUM OF THE PHILIPPINES UNIVERSITY
Cavite Campus

Assignment of Technical Critic

This is to inform that the research manuscript entitled _____

submitted by:

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

with the degree _____ under the College of _____
has/have been duly assigned under your faculty for technical editing and review with respect to form,
styles, systems and standards adhered by the institution and the discipline.

Conforme:

Technical Critic's Signature over Printed Name

Date Signed: _____
(mm/dd/yyyy)

Approved by:

Area Coordinator's Signature over Printed Name

Date Signed: _____
(mm/dd/yyyy)

Note: Reproduce and distribute two (2) duly signed copies to Technical Critic and Area Coordinator

LYCEUM OF THE PHILIPPINES UNIVERSITY
Cavite Campus

Assignment of Statistician

This is to inform that the research manuscript entitled _____

submitted by:

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

with the degree _____ under the College of _____
has/have been duly assigned under your faculty for statistical analysis with respect to appropriate
measurement tools and techniques.

Conforme:

Statistician's Signature over Printed Name

Date Signed: _____
(mm/dd/yyyy)

Approved by:

Engr. Annalyn Romero
Area Coordinator Signature over Printed Name

Date Signed: _____
(mm/dd/yyyy)

LYCEUM OF THE PHILIPPINES UNIVERSITY
Cavite Campus

Title Proposal Defense Score Sheet

Proponent(s):

- A. _____
B. _____
C. _____
D. _____
E. _____

Degree: _____ Sem. A.Y. 20__ - 20__

College: _____

Name of Adviser: _____

Proposed Title:

CRITERIA					RATING	MAXIMUM		
RESEARCH TOPIC					Group Grade			
Title								
Thesis statement is clear, concise, complete and original						20%		
Rationale								
The purpose / reason of the study is well stipulated, argued and justified						20%		
Organization								
Presentation and development of ideas are <i>clear</i> and <i>logical</i>						20%		
Subtotal – Group Grade						60%		
ORAL DEFENSE			Individual Grades					
Presentation			A	B	C	D	E	10%
1. Time allotted for presentation are met								5%
2. The visual presentation exemplified ideas, <i>concisely</i> and <i>comprehensively</i>								5%
Delivery			A	B	C	D	E	30%
1. Presenter is well prepared, appeared relaxed and confident								10%
2. Presenter is able to communicate effectively the ideas								10%
3. Exemplified mastery and reasoning ability in defending his/her proposal/section								10%
Subtotal – Individual Grade								40%
TOTAL								100%
(Please add the group grade [60%] to the individual grades [40%])								

FINAL RECOMMENDATION:

- The title proposal is accepted: ☐ **without** revision; 100% acceptability
☐ **with minor** revision(s); 80-99.99% acceptability; *refer to evaluation sheet*
☐ **with major** revisions; 75-79.99% acceptability; *for re -presentation*
- The title proposal is rejected: ☐ below 75% acceptability; *refer to thesis adviser*

EVALUATOR: _____

Signature over Printed Name

DATE _____

LYCEUM OF THE PHILIPPINES UNIVERSITY
Cavite Campus
Research Proposal Defense Score Sheet

Proponent(s):

- A. _____
B. _____
C. _____
D. _____
E. _____

Degree: _____ Sem. A.Y. 20__ - 20__

College: _____

Name of Adviser: _____

Title of Research/Project Proposal:

CRITERIA	RATING					MAXIMUM
RESEARCH PAPER/PROJECT	Group Grade					
Content						40%
1. Thesis statement / research objective(s) are well stated and justified						10%
2. Significance of the Study/Rationale are well argued						10%
3. Extent and sufficiency of Literature Review and References cited						10%
4. Appropriate data and methodology used						10%
Organization						10%
Presentation and development of ideas are <i>clear, logical</i> and <i>exhibits high standards of scholarship</i>						10%
Novelty and Impact						10%
1. The research endeavor exhibits a degree of innovation, originality, and ability to contribute to the existent body of knowledge						5%
2. Its impact and benefit to the society or community						5%
Subtotal – Group Grade						60%
ORAL DEFENSE	Individual Grades					
Presentation	A	B	C	D	E	10%
1. Time allotted for presentation are met						5%
2. The visual presentation exemplified ideas, <i>concisely</i> and <i>comprehensively</i>						5%
Delivery	A	B	C	D	E	30%
1. Presenter is well prepared, appeared relaxed and confident						10%
2. Presenter is able to communicate effectively the ideas						10%
3. Exemplified mastery and reasoning ability in defending his/her proposal/section						10%
Subtotal – Individual Grade						40%
TOTAL (Please add the group grade [60%] to the individual grades [40%])						100%

FINAL RECOMMENDATION:

- The thesis proposal is accepted: ☐ **without** revision; 100% acceptability
☐ **with minor** revision(s); 75-99.99% acceptability; *refer to evaluation sheet*
☐ **with major** revisions; 65-74.99% acceptability; *for re -presentation*
- The thesis proposal is rejected: ☐ below 65% acceptability; *refer to thesis adviser*

EVALUATOR: _____

Signature over Printed Name

DATE _____

LYCEUM OF THE PHILIPPINES UNIVERSITY
Cavite Campus

Proposal Defense Evaluation Sheet

Proponent(s):

- A. _____
B. _____
C. _____
D. _____
E. _____

_____ Sem. A.Y. 20__ - 20__

Degree: _____

College: _____

Name of Adviser: _____

Title of Research/Project Proposal:

Comments, Evaluation and Recommendations:

EVALUATOR: _____
Signature over Printed Name

DATE _____

LYCEUM OF THE PHILIPPINES UNIVERSITY
Cavite Campus

Final Defense Score Sheet
(Engineering and Architecture)

Proponent(s):

- A. _____
B. _____
C. _____
D. _____
E. _____

Degree: _____ Sem. A.Y. 20__ - 20__

College: _____

Name of Adviser: _____

Title of Research/Project:

CRITERIA				Rating		Maximum	
Design/Prototype							
1.	Degree of Design / Level of Technical Complexity					30%	
2.	Safety , Functionality , & Workmanship					20%	
Manuscript (Substance)							
1.	Content					20%	
2.	Organization					10%	
Subtotal-Group Grade						80%	
Presentation and Defense		A	B	C	D	E	
1.	Mastery in the Presentation of the Research Findings						5%
2.	Clarity in the Presentation of the Findings, Conclusions and Recommendations						5%
3.	Articulate Responses to the Examiners’ Inquiries						5%
4.	Correct Demeanor and Dress Code for the Occasion						5%
Subtotal – Individual Grade							20%
TOTAL							100%
(Please add the group grade [80%] to the individual grades [20%])							

FINAL RECOMMENDATION:

- The research project is accepted: ☐ **without** revision; 100% acceptability
☐ **with minor** revision(s); 75-99.99% acceptability; *refer to evaluation sheet*
☐ **with major** revisions; 65-74.99% acceptability; *for re -presentation*
- The research project is rejected: ☐ below 65% acceptability; *refer to thesis adviser*

EVALUATOR: _____
Signature over Printed Name

DATE _____

LYCEUM OF THE PHILIPPINES UNIVERSITY
Cavite Campus

Final Defense Evaluation Sheet

Proponent(s):

- A. _____
B. _____
C. _____
D. _____
E. _____

_____ Sem. A.Y. 20__ - 20__

Degree: _____

College: _____

Name of Adviser: _____

Title of Research/Project:

Comments, Evaluation and Recommendations:

EVALUATOR: _____

Signature over Printed Name

DATE _____

LYCEUM OF THE PHILIPPINES UNIVERSITY
Cavite Campus

Certificate of Originality

I/We hereby declare that the research paper entitled:

is my/our own work and that, to the best of my/our knowledge and belief, it contains no material previously published or written by another person nor material to which to a substantial extent has been accepted for award of any other degree or diploma of a university or other institute of higher learning, except where due acknowledgement is made in the text.

I/We also declare that the intellectual content of this thesis is the product of my/our work, even though I/we may have received assistance from others on style, presentation and language expression.

Proponent's Signature

Proponent's Signature

Proponent's Signature

Proponent's Signature

Proponent's Signature

Attested by:

Research Adviser's Signature

Chairperson's Signature

Noted by:

Dr. Arnel M. Avelino
Dean's Signature

Date of Completion: _____
(mm/dd/yyyy)

LYCEUM OF THE PHILIPPINES UNIVERSITY
Cavite Campus

Certificate of English Editing

This is to certify that the research manuscript entitled:

--

submitted by:

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

for the degree _____ under the College of _____
has been read and found it thorough and acceptable with respect to grammar and composition
by the undersigned English critic.

English Critic's Signature over printed name

Contact # : _____

Date of Completion: _____
(mm/dd/yyyy)

LYCEUM OF THE PHILIPPINES UNIVERSITY
Cavite Campus

Certificate of Technical Editing

This is to certify that the research manuscript entitled:

submitted by:

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

for the degree _____ under the College of _____
has been reviewed and found it thorough and acceptable with respect to form, styles and
standards adhered by the institution by the undersigned Technical critic.

Technical Critic's Signature over printed name

Contact # : _____

Date of Completion: _____
(mm/dd/yyyy)

LYCEUM OF THE PHILIPPINES UNIVERSITY
Cavite Campus

Certificate of Statistical Analysis

This is to certify that the research manuscript entitled:

submitted by:

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

Proponent's Signature over Printed Name

for the degree _____ under the College of _____
has been tabulated and analyzed by the undersigned statistician with respect to appropriate
measurement tools and techniques.

Statistician's Signature over printed name

Affiliation: _____

Contact # : _____

Date of Completion: _____
(mm/dd/yyyy)

YCEUM OF THE PHILIPPINES UNIVERSITY
Cavite Campus

Adviser Monitoring Sheet

Title of Research/Project:

Proponent(s):

A. _____
B. _____
C. _____
D. _____
E. _____

_____ Sem. A.Y. 20__ - 20__

Degree:

Name of Adviser

Date of Meetings	Topic/Concern	Action Taken / Comment/Recommendations	Signature

LYCEUM OF THE PHILIPPINES UNIVERSITY

Cavite Campus

Budgetary Estimate

Title of Research/Project:

--

Proponent(s):

___ Sem. A.Y. 20__ - 20__

A. _____

Degree:

B. _____

C. _____

College:

D. _____

E. _____

Name of Adviser:

PARTICULARS	QUANTITY	COST
TOTAL		

GANTT CHART

--

Name of Adviser: _____

[illegible]

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LYCEUM OF THE PHILIPPINES UNIVERSITY
Cavite Campus
Routing Slip of Final Manuscript

Title of Research/Project:

Proponent(s):

- A. _____
- B. _____
- C. _____
- D. _____
- E. _____

____ Sem. A.Y. 20__ - 20__

Degree:

College:

Name of Adviser:

Approval of Final Manuscript

	Date Received	Date Released	Remarks/Signature
Adviser			
Panel 1			
Panel 2			
Panel 3			
Panel 4			
Technical Critic			
English Critic			
Research Teacher			
Vice President for Academic Affairs			

Distribution of Copies of Final Manuscript/Journal Article

	Received by	Signature
Dr. Anjerick Topacio Research Coordinator IEEE Journal Format & max 3-min Video Documentation (g-drive link)		
ARC (Hardbound copy)		
Research Office (soft copy)		
Vice President for Academic Affairs		

APPENDIX III

Source Documentation

Source: LPU Student Research Manual – Undergraduate (2014)

APPENDIX IV

Manuscript Setting

APPENDIX V
Thesis Policies and Guidelines

Thesis Policies and Guidelines

1. General Requirements

Students must have passed all 1st to 3rd year (4th year for BS Architecture) subjects before working on a thesis project. All professional subjects must be taken prior to thesis, in case that there are professional subjects not taken yet, the said subjects must be taken simultaneously with thesis / design project.

Degree	List of Professional Subjects
BS Architecture	Architectural Design 8 Research Methods for Architecture Practicum/ OJT
BS Civil Engineering	Highway Engineering Engineering Economy Structural Theory Construction Material & Testing Hydraulics Hydrology Building Design Geotechnical Engineering Transportation Engineering Water Resource Engineering Computer Aided Drafting & Design Practicum
BS Computer Engineering	Object Oriented Programming CPE Drafting & Design Software Engineering Database Management Digital Signal Processing Advanced Logic Circuit Design Data Communication & Networking Control System Discrete Math Engineering Management Practicum
BS Electronics Engineering	Energy Conversion Microprocessor System Numerical Method Discrete Math Digital Communications Engineering Management

	Feedback and Control System Industrial Electronics Fundamentals of Material Sciences Signals, Spectra & Processing Electromagnetics Practicum
BS Electrical Engineering	Electrical Trans & Distribution Engineering Economy AC Machinery AC Apparatus & Devices Industrial Power Electronics DC Machinery Methods of Research Electrical Circuits 3 Logic Circuit & Switching Principles of Communication Control Systems Analysis Numerical Methods Electromagnetics Practicum
BS Industrial Engineering	Methods of Research Systems Engineering Production Systems Operation Research Managerial Accounting Advanced Ergonomics Marketing Systems Industrial Quality Control Engineering Economy Industrial Organization Management Industrial Material & Processes Practicum
BS Mechanical Engineering	Safety Engineering for ME Engineering Management Methods of Research Refrigeration Systems Machine Design and CAD Combustion Engineering ME Laboratories DC & AC Machinery Heat Transfer Fluid Machinery / Fluid Mechanics Machine Elements

	Materials of Engineering & Testing Computer Aided & Drafting Practicum
BS Computer Science	Research Methods in Computing Software Engineering 2 Automata and Language Theory Information Assurance and Security Quantitative Methods
BS Information Technology	Research Methods in Computing Quantitative Methods Information Assurance and Security 1 Application Development and Emerging Technologies
BS Library and Information Science	Introduction to Library and Information Science Collection Management of Information Resources Information Resources and Services 1 Information Resources and Services 2 Organization of Information Resources 1 Organization of Information Resources 2 Indexing and Abstracting Management of Libraries and Information Centers Information Literacy Library Materials for Children and Young Adults Introduction to Records Management and Archives

2. **Students are expected to**
 - a. Apply appropriate concepts, principles, methods, and techniques of their chosen field of study
 - b. Organize and work in a team, sharing resources, tasks and responsibilities.
 - c. Follow rules and procedures, a plan of activities to meet pre-determined output deadlines.
 - d. Organize, present and write professional quality reports acceptable to the faculty of the College of Engineering, Computer Studies and Architecture.
 - e. Contribute to the research thrusts of the COECSA.

3. The **main requirement** for Research 1 / Feasibility Study 1 / Capstone 1 (ARDES09, CERL01E, CPRL01E, EXRL01E, EERL01E, IERL01E, PFSL01E, MERL01E, LISN12L, CPTN01C, THEL01C) is the submission and defense of the project proposal. Submission to the adviser of the complete and approved draft for the first three chapters of the thesis, incorporating suggestions and comments of the panel of examiners.

4. The **main requirement** for Research 2 / Feasibility Study 2 / Capstone 2 (ARDES10, CERL02E, CPRL02E, EXRL02E, EERL02E, IERL02E, PFSL21E, MERL02E, LISN13L, CPTN02C, THEL02C) is the final thesis defense and the submission of prototype (if applicable), hardbound and soft copies approved by the Dean and the Vice President for Academic Affairs.
5. **Starting Out**
 - a. Two (2) to maximum of four (4) students shall work on a thesis topic. Students must inform their research teacher of the thesis grouping not later than two (2) weeks after the start of the Methods of Research class.
 - b. Upon the instruction of the research teacher, the groups shall identify potential topics and submit it using the appropriate template for approval and evaluation of the research teacher.
 - c. As scheduled, the group will be scheduled for a title defense of the approved topics at the end of the term for Research Methods.
 - d. The students' thesis adviser will be assigned by the research teacher after the title defense.
 - e. Students enrolled in the research methods subject must attend the Research Orientation scheduled by the College.
6. **Preparing for Defense**
 - a. Except in meritorious cases, the panel of examiners for the final thesis defense shall be identical to that of the thesis proposal defense. The research teacher shall submit a list of proposed examiners at least two weeks before the schedule of thesis defense.
 - b. Thesis proposal / final thesis defense may only be scheduled after the students get the approval of their thesis adviser, demonstrated by having the adviser's signature on the Endorsement and Request for Defense.
 - c. It is the responsibility of the students to coordinate with the examination panel and their thesis adviser regarding defense schedule, submission of paper and other matters relating to the thesis. In scheduling their defense, the research teacher is expected to consider the availability of the advisers and all examiners. The schedule will be forwarded through a memo approve by the Dean.
 - d. The thesis manuscript must be submitted at least one (1) week before the scheduled oral defense to the examiners. The examiners shall sign in the endorsement and request for defense to acknowledge receipt of the paper. Failure to submit the manuscript will not be entertained during the oral defense.
 - e. Students must be punctual for their thesis proposal / final thesis defense. Failure to appear on time could mean a deferment or a failing mark. Presentation of the thesis proposal / thesis must be done in PowerPoint or similar software. Students are expected to be in corporate attire during their defense.

7. Oral Defense

- a. A thesis proposal / final thesis defense may push through even if one of the examiner is not present, provided that the thesis adviser and the chairman of the panel are present. The absent examiner is however required to submit his/her comments on the proposal / thesis to the chairman of the panel.
- b. Thesis proposal defense shall have a maximum duration of one hour, 20 minutes of which is allotted for presentation. The thesis adviser shall gather comments and suggestions of the panel and provide a copy to the thesis proponents. He/she shall also keep a copy of the proposal and comments, which shall be considered as the official copy. Official copy may be deposited in the department chair's office for safekeeping.
- c. Final defense shall have a maximum duration of two hours of which is allotted for presentation. The thesis adviser shall gather comments and suggestions of the panel, and provide a copy to the students to guide them with their revisions.
- d. If there is a prototype, testing and checking of the prototype shall be done before the oral presentation. The prototype must be working properly and tested accordingly before proceeding to the oral defense.

8. Output, Grades, and Awards

- a. Thesis proposal and thesis should be written following the LPU style guide.
- b. Comments and suggestions by the panel must be incorporated in the final version of the thesis. The students must accomplish the Routing Slip until it is signed by the Research Teacher before binding the thesis.
- c. As a final requirement, students must submit the hardbound copy, soft copy and other requirements stated in the routing slip.
- d. The accomplished routing slip will be checked during the clearance period for the graduating students.
- e. Grade for thesis proposal / final thesis defense shall be the average of the grades given by the panel members, and released by the research teacher only upon completion of all requirements. Students will however be informed immediately after the defense whether they passed the defense or not.
- f. An Incomplete grade may be given to students for non-completion of requirements after the oral defense. The group, however, may advise the research teacher in advance if they cannot make it in the defense. An incomplete grade will be given in this case. Incomplete remarks may however be given only once for each enrolment.
- g. In case of re – defense, the group will be given at most two (2) weeks to comply and present again following the recommendations of the panel.
- h. Non-appearance in the scheduled re-defense, or non-compliance in the recommendation of the panel, means failure in the final term grade.
- i. The panel of examiners, research teacher, and the program chair may recommend the students for the College award for BEST THESIS. The nominees for BEST THESIS will present in the College Research Colloquium scheduled by

the college. The BEST THESIS award will be given during the College Research Colloquium.

9. Intellectual Property

The COECSA shall have the right to use and reproduce for non-commercial purposes any new products, procedures and results that may be derived from thesis research done by students under the mentorship of a faculty member of the department.

10. When Problem Arise

- a. If students want to transfer from one research group to another, they must immediately inform their research teacher in writing. If a thesis group decides to split up, students concerned must also immediately inform the research teacher in writing.
- b. In cases when an adviser cannot continue to advise a thesis group, the research teacher shall assign a new adviser to the students.
- c. Any revision to the approved proposal must be submitted in writing to the panel of examiners for approval. Rationale for the revision must be provided by the students and approved by the adviser.
- d. Questions or complaints by students or faculty about the conduct of thesis, thesis defense or thesis grade shall be referred to the Program Chairperson.
- e. Matters not covered in the policies and guidelines shall be referred to the Program Chairperson/Research Coordinator for final solution.

APPENDIX VI
Journal Article Format

A Proposed Agritourism Farm with Learning and Training Center in Talavera, Nueva Ecija

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Abstract—Agricultural advancement in the country is still behind due to lack of proper learning and training centers for agricultural training, decreased farm income, and limited opportunity for agricultural innovations, development, and technology. With these, this paper presents analysis of data, architectural programming, conceptualizations, and information, that aims to develop an architectural design for an agritourism farm with an agricultural learning and training center. Data gathering methods, including review of studies and literature, surveys, interviews, site visitations, site analysis, and visitation of related projects were conducted to ensure that the project output adhered to the goals and objectives of the study. After examining the data collected, the researchers inferred the following: farms can adapt agritourism to boost farm income and maintain farms sustainably; agricultural learning and training centers must maximize open spaces and provide the needed spaces for farm demonstrations and practices following the guidelines set by the Agricultural Training Institute (ATI) and; the importance of using locally sourced materials and passive design approach on each facility to provide rural lifestyle experience. In an effort to meet these needs architecturally, the study integrated an agritourism farm with a learning and training center using nature - integrated design approach.

Index Terms—Agriculture, Agritourism, Learning and Training Center, Post – harvest Facilities, Nature – Integrated Design

I. INTRODUCTION

AS a country known for its agricultural production, the Agricultural development of the Philippines still takes time due to stagnant adaptation of technological innovations and modern mechanizations in farming [1]. In addition, decreased income and profit of farmers had gradual impact on their daily lives and productivity as well. In an attempt to address these issues, the Farm Tourism Development Act of 2016 (R.A. No. 10816) strengthened the promotion and development of farm tourism in the country [2]. To maximize the purpose and use of farms, owners can utilize it as a learning center.

With the guidelines and requirements set by the

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Agricultural Training Institute (ATI) and R.A. No. 10816, the facilities are designed to provide facilities suitable to the needs of both farmers and tourists [3]. Situated in Talavera, Nueva Ecija, it is located in the agri – industrial zone, which is allotted for promotion and establishment of agri – businesses in the area. To improve agricultural development, the facilities were designed for both indoor and outdoor trainings and demonstrations, ensuring safety and efficiency through its space planning [4]. Nature – integrated design, a framework that connects humans with nature through the built environment, can be utilized to build connection between users and the environment and promote sustainable tourism [5].

Through architecture, the study aims to propose an agritourism farm with a learning and training center using nature – integrated design to support the agricultural initiatives of the aforementioned departments.

II. PROJECT GOALS AND OBJECTIVES

A. Project Goals

With the continuous decline and problems in the agricultural industry, the goal of the proposed project is to provide proper and better facilities for farmers in order to adapt with the technological advancements and machinery emerging today and in order to promote local products and the rural lifestyle. The goal of this study is to design an agritourism farm with a learning and training center that will support and provide their needs and may elevate their income.

The study will also meet the following specific design objectives:

- 1) To propose an agritourism farm with a learning and training center, aligned with laws and programs such as Republic Act No. 10816, also known as the Farm Tourism Development Act of 2016, and with accordance to the requirements of the Agricultural Training Institute (ATI) to help the community on adapting with modern technologies and machineries used in agriculture and market industry.
- 2) To design better facilities for farm training, workshops, seminars, and other related activities, and to cater the needs of farm owners and farmers, to perform their functions collaboratively.
- 3) To provide effective and good space programming based on systematic study conducted by the researchers.

- 4) To provide an agritourism farm for leisure of visitors, to offer rural lifestyle and experience, and boost the value and image of the agricultural sector.

III. RESEARCH METHODOLOGY

A. Research Design

The researchers employed both qualitative and quantitative research techniques for the study.

The goal of the qualitative research approach is to gather non-numerical data in order to comprehend phenomena, behavior, and perceptions. It is used in the study through the examination of several associated works of literature, discussions with experts, staff members, and other pertinent persons, as well as data collection and recording from observations of relevant structures, visits to related sites, and user behavior analysis.

The quantitative research method makes use of numerical or statistical data to interpret observations and examine relationships and patterns. It is used in the study by asking users of the Hall of Justice about the facility's human-centered design criteria.

B. Data Gathering Process

- 1) *World Wide Web* – the World Wide Web was utilized for data collection, review of pertinent international projects, and communication with various government agencies. The benefits of using WWW to collect behavioral data include instant access to a large number of possible respondents and previously hidden populations, respondent openness and full engagement, student research opportunities, and decreased research expenses.
- 2) *Library* – the purpose of visiting libraries was to obtain more pertinent knowledge on relevant books. Libraries have extensive resources in the form of books, journals, periodicals, articles, and bibliographies, and play a crucial role in research. A review of unpublished or published research, such as master's or bachelor's theses, linked to this project proposal or architecture was conducted by visiting the libraries of other institutions.
- 3) *Government Organization / Agencies* – Agencies and organizations related to the study were visited to gather information. These include the Municipal and Provincial offices of the Department of Agriculture and Department of Tourism, PhilRice Institute, and Agricultural Training Institute (ATI).
- 4) *Key Resource Person Consultation* – Gather information or data through discussions with resource persons who have knowledge in the planning and design of farms, who operate a farm tourism business, and who work in the agritourism industry in order to acquire technical data that cannot be derived from written documents.
- 5) *Survey* – The purpose of the questionnaire is to learn more about users' expectation on the experience and amenities to offer by the farm. It also identifies the nature – integrated design elements that are included into the architecture of the building. The relevant research and studies that are examined in this study are the source of the nature – integrated design parameters.
- 6) *Review and Visitation of Related Projects* – Visits,

studies, and reviews of related projects were conducted in order to establish a standard for the probable techniques in space derivations and sizing and to conduct a qualitative examination of its form and function. As a possible foundation for concepts, the visitation of any connected initiatives, as well as their strengths, will be considered, as will their weaknesses.

IV. RESULTS AND ANALYSIS

A. Data from Interviews

From the interviews, the researchers were able to gather data about programs, services, amenities, and spaces offered and present in agritourism centers.

B. Data from Visited Related Projects

By visiting related projects, the researchers recorded the departments, offices, and spaces within an agritourism farm and learning and training centers. They also examined each facility to document issues present such as poorly – design spaces, insufficient facilities, poor zoning of spaces, and poor landscaping.

C. Site Analysis

The goal of the study is to identify a location that is both suitable and workable for the suggested project. The researchers chose three Talavera barangays as prospect sites in order to achieve this. The project will be able to successfully support functions and activities and positively impact both the surrounding community and users at the site that is assessed to be most suitable. The prospect sites need to be analyzed using the site criteria that were developed during the study in order to ascertain this.

TABLE I
SITE CRITERIA AND WEIGHT FACTOR FOR MUNICIPALITY

Criteria	Rating	Discussion
Agricultural Production	35%	Because the project focuses on agriculture, the municipality must be among the top crop producers in order to maximize its resources.
Tourism	25%	The number of tourists is the second priority as it is one of the major objectives of the project. It is vital to attract a number of tourists for the success of the project.
Proximity	20%	Proximity to nearby related facilities is an advantage to the project.
Competitiveness Index	10%	Local government authorities can use the Index as a diagnostic tool to evaluate the competitiveness of their town or city and to pinpoint opportunities for improvement and teamwork. Data can offer understanding for determining

		policies, planning for development, and encouraging investment
Population	10%	High number of residents shows that the municipality is being developed. Being populated offers its people new opportunities because of the project.
Man – made and Natural Hazards	5%	To avoid endangering the project and its users, it is crucial to have a secure site that is free of both natural and man-made risks.

TABLE II

SITE CRITERIA AND WEIGHT FACTOR FOR SITE

Criteria	Rating	Discussion
Proximity	25%	Proximity of the site to nearby related facilities and tourist spots is advantageous. It is the top priority for choosing the site.
Utility Availability	20%	To use the project properly, the site needs to have access to telecommunications, sewage, energy, and water. It is vital to the project, especially the irrigation system.
Accessibility and Security	15%	The site must be accessible through various road networks for users' easier access to it and must be secured for the safety of the users.
Zoning and Local Ordinances	15%	Zoning must be taken into account because it is one of the criteria for selecting the ideal site.
Site Features	10%	The site's features and characteristics will influence planning and development, thus picking the right location is crucial.
Man – made and Natural Hazards	5%	To avoid endangering the project and its users, it is crucial to have a secure site that is free of both natural and man-made risks.
Topography and Existing Vegetation	5%	A site's topography has an impact on a number of variables, including surface water flow, erosion and sedimentation, and vegetative development. These elements are crucial for site planning, which includes identifying and protecting environmentally sensitive areas.

Site Potential 5% The site is a potential area for a related project development.

1) Prospect Sites

a.) Site 1 – Brgy. Sampaloc, Talavera, Nueva Ecija



Fig. 1: Site 1

Source: Google Earth

b.) Site 2 – Brgy. La Torre, Talavera, Nueva Ecija



Fig. 2: Site 2

Source: Google Earth

c.) Site 3 – Brgy. San Miguel na Munti, Talavera, Nueva Ecija

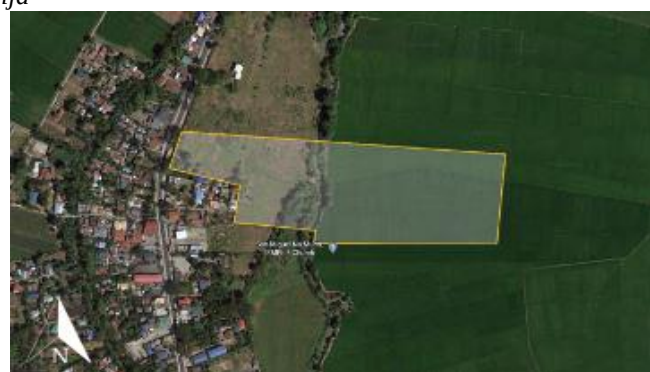


Fig. 3: Site 3

Source: Google Earth

TABLE III

SITE CRITERIA RATING OF PROSPECT SITES

Criteria	Rating	Sites		
		Site 1	Site 2	Site 3
Proximity	25%	5	4	2
Utility Availability	20%	5	4	5
Accessibility and Security	15%	5	5	3

Zoning and Local Ordinances	15%	5	5	5
Site Features	15%			
a. Lot Area		5	4	2
b. Soil Condition		5	5	5
c. Geometry of Site		5	5	2
d. Expansion		5	5	3
Man – made and Natural Hazards	5%	3	3	3
Site Potential	5%	5	4	2
TOTAL		4.8.	4.4	3.2

C. Site Profile and Inventory of Chosen Site

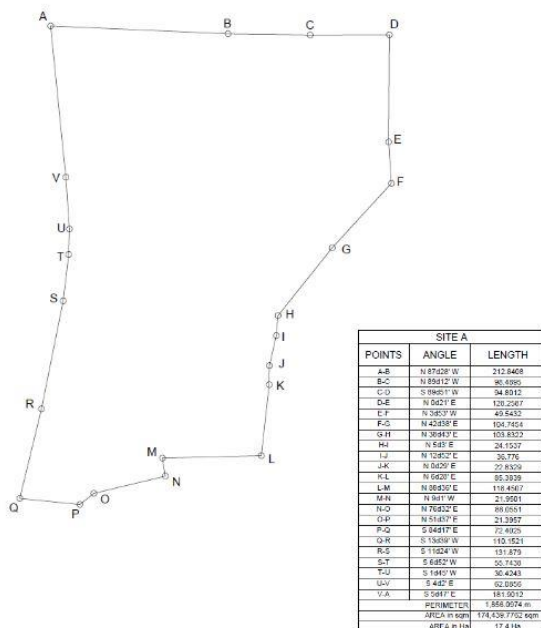


Fig. 4: Site's Bearing

a) Movement and Sensory

The site as seen in the figure illustrates the traffic on the nearby Talavera Bypass Road and Maharlika Highway. While Talavera Bypass Road has low to moderate vehicular traffic, Maharlika Highway has low to moderate vehicular traffic. The agricultural land is visible around the site. Because it is not directly beside the Maharlika Highway, noise pollution will not be an issue.



Fig. 5: Site's Road Networks

b) Slope, Vegetation, and Existing Structures

The site has a flat surface with various numbers of existing trees. There are no existing structures present in the site. A small private farm and a private resort is located around the site. Also, the NIA Irrigation Canal is directly beside (left side) of the chosen site located at Brgy. Sampaloc, Talavera, Nueva Ecija.



Fig. 6: Site's Nearby Irrigation Canal

c) Sun Path Analysis

The researchers can strategically add skylights and shading devices to the proposed buildings to maximize natural light, reduce reliance on artificial lighting, and create comfortable and aesthetically pleasing indoor settings by knowing the sun's path, which helps them determine which portions of the building will receive direct sunshine, indirect sunlight, or what will remain shaded at certain times of the day.



Fig. 7: Sun Path Orientation of the Site

d) Prevailing Wind Analysis

Information on the year-round predominant wind directions and speeds is provided by wind path analysis. Designers can utilize this information to determine which building orientation will maximize natural ventilation. In addition, windbreaks, such trees or bushes, can be thoughtfully placed on the site's north-east and south-west faces to create sheltered microclimates by researching wind patterns.

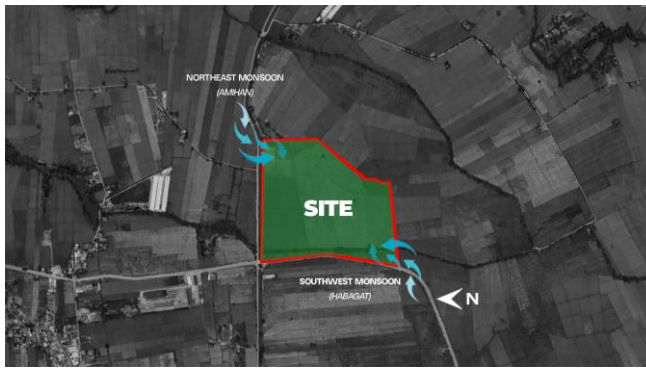


Fig. 8: Wind Analysis of the Site

e) View Analysis

The site is surrounded by rice fields and some commercial areas. The property is close to the town of Talavera, which has a variety of commercial and industrial areas, tourist destinations, and natural landscapes. On the north - east, eastern, and southwest view of the site, Mt. Sawi of Gabaldon, Nueva Ecija, Mt. Mabilog of Pantabangan, Nueva Ecija, and Mt. Mingan of Dingalan Aurora, can be seen.

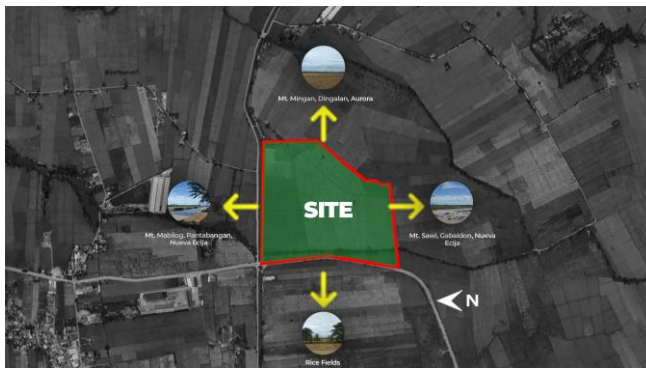


Fig. 9: View Analysis of the Site

f) SWOT Analysis

Strengths	Weaknesses	Opportunities	Threats
• The site is directly accessible by a main road (Maharlika Highway) and a bypass road (Talavera Bypass Road). • Directly beside the National Irrigation Administration (NIA) Irrigation Canal. • The site is near to various commercial establishments such as Waltermart and Savemore Talavera, the National Irrigation Authority	• The site's natural grade line (NGL) is lower than that of the major Road. • The site has limited streetlights as the bypass road was newly developed. • The site is mostly accessible via tricycles and private vehicles. • Large trucks pass through the Bypass road that may cause slow vehicular	• The site is # hectares and is open for wide expansion in the future. • It is near local commercial establishments, local eateries, and other amenities, making it user - friendly. • It can increase the municipality's economic growth. • The presence of	• Flooding can occur in times of typhoons. • The weather might be hot when it is summer due to extreme weather conditions. • Water shortage / drought may be encountered because of the extreme weather conditions. • Possible emergence of farm competitors.

- Talavera Office, Talavera General Hospital, Talavera Fire Station, Talavera Municipal Hall, schools, and other tourist spots such as resorts.
 - It is accessible to utilities: electricity, water, sewage, and telecommunications.
 - The soil quality is suitable for crop production.
 - The site has a vast field perfect for farming demonstrations.
- possible two entrances and exits makes it easier for users to get into the project and lessen the traffic it may cause.

IV. ARCHITECTURAL DESIGN TRANSLATION

The study has identified agricultural and tourism – related issues which can be addressed through architectural design responses through nature – integrated design approach. The design and plan for an agritourism farm with a learning and training facility that will architecturally address the previously identified issues is presented by the researchers.

A. Design Goal

The primary goals of the proposal are to design and provide an architectural design solution that will boost the tourism and agricultural growth of Talavera, Nueva Ecija, and will cater the needs of farmers to improve in the agricultural industry. Also, the project will be a catalyst of showcasing agriculture's importance in the country and how rural lifestyle can be experienced. This will also benefit the surrounding community in Talavera, Nueva Ecija.

B. Design Objectives

The design of the project intends to achieve the following:

- 1) To design spaces where people can learn, train, and relax comfortably, effectively, and securely.
- 2) To design an agritourism farm that will make the users feel and appreciate rural lifestyle and agriculture.
- 3) To design an agritourism farm that will encourage the community to engage more in agriculture and other agricultural related activities
- 4) To design a learning and training facility that will boost the morale of farmers and will engage them to participate more in various training.

C. Design Philosophy

"Nature should be used as a guide and that the indigenous flora should be respected instead of altered." – Frederick Clements

D. Design Theories and Paradigm

The main design concept of the project revolves around nature – integrated design. It highlights the advantages of natural components, like the importance of sunlight, fresh air, lush vegetation, and picturesque vistas. Consisting three (3) kinds of experience of nature: direct experience of nature, indirect experience of nature, and experience of space and place, were incorporated to the project. The proponents considered visual connection and access to nature, natural light and ventilation, and appropriate material choices, upon planning and designing the proposed project. One of the project's design goals is to blend the development with its environment in order for it to not be a distraction but to be a part of nature itself.

The project aims to apply passive design approach following nature – integrated design and vernacular architecture. With the fusion of the design characters of *bahay kubo* and old Filipino houses, it offers rural experience and lifestyle to its users, dissociating from the busy urban life.

E. Design Considerations

- 1) *Accessibility*
One of the main aspects that must be used in public facilities is that the general design must be friendly to individuals with disabilities.
- 2) *Acoustics*
Controlling noise pollution is a part of creating a building environment with acceptable acoustics. The users of the complex must be relaxed without the outside noise. Building occupant and system noise can travel throughout the structure and have an impact on its functionality.
- 3) *Aesthetics*
The design shall perform its functions to satisfy the needs of its users. Having aesthetically pleasing design also catches the interest of the public.
- 4) *Anthropometrics*
Instead of forcing individuals to conform to the building's design, buildings should be made to accommodate the human body. An anthropometric approach to architecture aims to provide the greatest level of comfort for each individual.
- 5) *Compliance to Ordinances*
The proposed project must follow the building laws as it protects, secures, and generates welfare for the building and its occupants.
- 6) *Circulation*

To develop an effective and well – organized flow that is specific to the user's functions, the design must separate primary from secondary users.

7) *Energy Efficient*

A design must include the methods and strategies that will reduce the cost of energy on a daily basis. Energy efficiency has a number of advantages, including a decrease in greenhouse gas emissions, a decrease in the need for imported energy, and a decrease in the price of electricity.

8) *Ergonomics*

The building must be designed for the comfort of the users. By factoring in human skills and limits, such as body size, strength, skill, speed, sensory capacities (vision, hearing), and even attitudes, ergonomics strives to create safe, comfortable, and productive workplaces.

9) *Expansion*

The use of planning components enables the structure to quickly adjust to required reductions or expansions without compromising facility functionality.

10) *Innovation*

With the continuing development and emerging innovations, making use of it can help to lessen financial maintenance on systems. Along with the farm mechanization programs are small innovations that may help in the farm systems of the project.

11) *Material Sustainability*

This relates to all issues with the efficient use of resources and the choice of environmentally friendly materials.

12) *Safety and Security*

Security systems are provided in each building and around the area to ensure safety of all visitors and staff that will enter and work to its users.

13) *Technology and Trends*

It is a must to keep up with the emerging technology and trends as it provides faster and even more convenient work to its users.

14) *Utility*

Utilities are vital to the project as it is one of the primary backbone for it to work. The development must have proper utility to have more efficient and cost – effective development.

F. Presentation of Drawings

1) Master Development Plan



Fig. 10: Entrance perspective



Fig.11: Entrance perspective

2) Reception and Administration Building

The reception area is located near the drop-off of the users. It has an open plan to make the users feel more closely to nature and to have natural ventilation. It has a walkway connected to the restaurant then connecting to the various spots in the farm.



Fig. 12: Reception exterior perspective



Fig. 13: Admin Building 1 exterior perspective



Fig. 14: Accounting Office interior perspective

3. Restaurant

One of the main amenities and services offered in the project was the farm – to – table dining in the restaurant. The restaurant is placed near the reception and administration building almost at the center of the development to make it accessible for public use. It has two levels and an outdoor dining area to cater more users. Circular in plan, the restaurant aims to maximize the view of the vast field around it. The roof was inspired from the *palay* grains linked together in a circular manner as well.



Fig. 15: Restaurant exterior perspective



Fig. 16: Restaurant interior perspective

4) Learning and Training Center

The spaces were based on the requirements of the Agricultural Training Institute (ATI). It has a lobby with an open atrium, clinic, locker rooms, restrooms, conference room, offices, enclosed learning rooms, storage room, electrical and utility room, maintenance room, and an open outdoor training area intended for outdoor demonstrations. The learning and training center is located near the farmer's dormitory. The roof design of the facility was inspired from the *palay*'s form and husk when laid on the ground. Some parts of the roof were connected to the ground as it symbolizes how rice grew from the ground. It also acts as support for the roof of the facility. The design also intends to

blend within the field as the roof was design in an organic form.



Fig. 17: Training Center exterior perspective



Fig. 18: Learning Room interior perspective



Fig. 19: Trainers' office interior perspective

5) Accommodation Rooms

The accommodation room provided has two types: Type 1 can accommodate 2 people while type 2 can accommodate 4 – 6 people. The shape of the plan originated from the oblongate shape of rice grain and was softened on both of its edges. While, the roof was inspired from the rice husk. For the design elements, similar to old houses present in the province, the railings were design based on common railing designs used in old Filipino houses as well as using *callado*.



Fig. 20: Accommodation Rooms exterior perspective

6. Farmer's Dormitory Rooms

The dormitory is located near the learning and training center for farmers' easier accessibility. With a common dining space and kitchen, it has 12 rooms that can cater 144 farmers. The design is similar to the accommodation rooms to impose equality among users. All dorms were connected with a walkway planned inspired from rice stalks.



Fig. 21: Farmer's Dormitory Room exterior perspective

7) Farmer's Dormitory Dining Hall

The dining hall is placed near the dorm kitchen and training center. It has a circular open plan for easier accessibility, better ventilation and to maximize its function.



Fig. 22: Farmer's Dormitory Room exterior perspective

8) Multi – Purpose Hall

Located near the reception building, it serves as an additional area open to public use for events. It has its own control room, storage room, restroom, and food preparation area. It is designed aligned with passive cooling techniques allowing cross ventilation within the hall. The jalousie window allows both natural light and air to pass through the hall. It has pivotal doors to maximize the view and function of the hall.



Fig. 23: Multi-purpose Hall exterior perspective

9) Open Market

Located near the parking area and reception, the market allows farmers and other businesses to sell their products to tourists and locals. It allows natural light and air to pass through. It would be easier to trade goods as the market has an open plan design.



Fig. 24: Open Market interior perspective

10) Spa

Located in the middle of the field close to the reception building, the spa offers relaxation services both indoor and outdoor. It has a main reception building and another building for the spa services itself.



Fig. 25: Spa Reception exterior perspective



Fig. 26: Spa Reception interior perspective



Fig. 27: Spa Massage Room interior perspective

11) Information Resource Center (Library) and Community Center

To offer and maximize the library, it also functions as a community center for the locals. The design of the library and community center was based on the circular forms following the straw hats used by farmers in the field. By splitting it into two parts, a higher roof was placed in order to provide better ventilation within the facility.

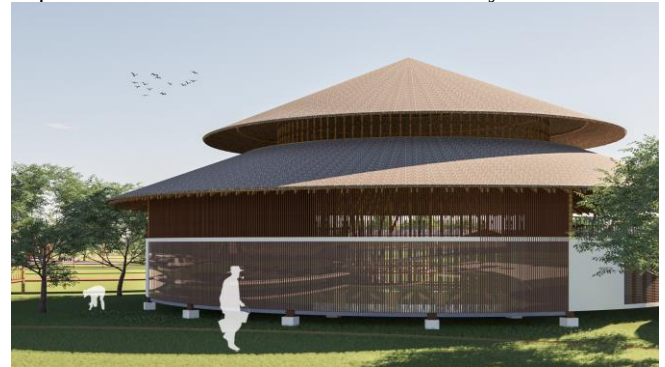


Fig. 28: Library and Community Center exterior perspective



Fig. 29: Library and Community Center interior perspective

12) Outdoor Cottages

These cottages are situated along the rice field linked through walkways. Each cottage can accommodate at least 6 – 12 people and its design was a mix of bahay – kubo style and the old historical houses found in Nueva Ecija.



Fig. 30: Outdoor Cottages exterior perspective

13) Drying Warehouse

It houses flatbed dryers and was designed with vertical louvers around it. It is accessible through 4 wide doors that can fit tractors for easier transfer of grains.



Fig. 31: Drying Warehouse interior perspective

14) Crop Storage

The storage was designed based on the standard guidelines of the Philippine National Standards for Agricultural Structures – Warehouse for Bag Type Storage of Grains. The warehouse can accommodate up to 100,000 cavans of rice.



Fig. 32: Crop Storage interior perspective

15) Equipment Warehouse

It caters tractors used in the farming process to protect it from extreme heat, rain, and for its security as well.



Fig. 33: Equipment Warehouse exterior perspective

VI. CONCLUSION

The study proposed architectural responses for the agricultural and tourism related issues aforementioned such as slow agriculture development progression, poorly – design spaces, lack of post – harvest facilities, and decreased farm incomes and number of farmers.

To keep up with agricultural advancements, farmers need to undergo ongoing trainings such as giving them the skills and information they need to properly carry out and use farming techniques and practices [6]. Farmers gain a thorough awareness of wide range of abilities, methods, and domains that have an impact on agricultural operations thanks to continuous training. It familiarizes aspiring and new farmers with the fundamentals of managing small to large farms [7].

For Filipino farmers, a persistent issue has been the lack of storage facilities, farm-to-market routes, and equipment needed to minimize crop losses from exposure, pests, and natural deterioration. Most farmers are unable to turn a profit when these facilities and tools are lacking since these losses destroy a significant number of crops before they are sold. Postharvest facilities (PHFs) allow rice farmers to decrease postharvest losses and produce high-quality milled rice using modern technology, which can increase rice farmers' profitability [8].

[9] The agricultural sector receives the lowest hourly wages when compared to those in the industrial and service sectors; as a result, the agricultural sector—also known as "a poor man's sector"—has the highest concentration of poverty in the nation. Many children of agricultural laborers have already left their fields in search of better paying jobs in the cities due to the association between agriculture and poverty. Consequently, there is a declining pool of agricultural laborers and a low regard for agriculture among the younger population [10].

Through related studies and visitation of related projects, the researchers surveyed the existing condition of farms and learning and training centers and discovered that most of the facilities have poorly – designed and incomplete spaces. The researchers ensured to adhere to nature – integrated design principles for the architectural translation of the study.

The output of the study, an architectural design of an agritourism farm with a learning and training center through nature – integrated design, was conceptualized and formulated in accordance with the aforementioned problems and guidelines.

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BIOGRAPHY



Ms. XYZ was born on May 31, 2000 in Luisiana, Laguna. She began her pursuit of her dream to be an architect by obtaining a Senior Highschool diploma in Science, Technology, Engineering, and Mathematics in

Lyceum of the Philippines University-Cavite International School, graduating with honors. She went to Lyceum of the Philippines University-Cavite to take up Bachelor of Science in Architecture from 2018 to 2024. During her college days, she was affiliated in United Architects of the Philippines Student Auxiliary - LPU Chapter from 2019 to 2020 as Chapter Auditor and Chapter Treasurer from 2020 to 2021. She was then affiliated to United Architects of the Philippines Student Auxiliary Area B from 2021 to 2022 as the Marketing Directress for Area B. She also served as the National Chair, Committee on Chapter Organization in United Architects of the Philippines Student Auxiliary Fiscal Year 2022 to 2023.

Additionally, she also finished her internship in Arkiworx Design Build and CADD Services in 2023 where she

worked on providing assistance to the Junior Architects with the supervision of Licensed Architects.

During her senior days in the university, she was recognized as a Dean's Lister student. She also participated in the HCG ArchiNEXT Young Designer's Design Competition 2022 where she got a chance to be a Magic 32 finalist. Along with her thesis partner, she was able to secure the Top 10 in the Architectural Theses of LPU-Cavite.



Mr. XYZ was born on February 28, 1999 in Trece Martires City, Cavite. Upon completing her senior high school in 2018 with honors, she began to pursue her BS Architecture education in Lyceum of the Philippines University - Cavite

(LPUC). She served several organizations in the university. As part of her passion, she became a member and further on became the President of the LPU Cavite Dance Company in 2018 - 2019. She is an active member of the United Architects of the Philippines Student Auxiliary – LPUC Chapter where she served as the Chapter Commission on External Affairs in 2019 - 2020, Chapter Secretary in 2020 - 2021, and Chapter President in 2021 - 2022. Aside from it, she also served as the Business Manager for the College of Engineering, Computer Studies, and Architecture Student Council in 2020 – 2021. Exploring more of her potential, she entered the UAPSA National level where she was appointed to be the District Director for District B3 in 2022 – 2023 and then the National Executive Director for External Affairs in 2023 - 2024.



Arch. XYZ is a Minister, an architect, an instructor, a father to four and a husband to one. Born in Indang, Cavite on November 25, 1986. Graduate of Architecture Technology at TUP Cavite and BS Architecture at

Manuel L Quezon Univeristy.

VIDEO DOCUMENTATION:

https://drive.google.com/file/d/1RHe2MXVlRjo_Lc8wJE0yZRnCXlXqpRaX/view?usp=sharing